



DATA DESCRIPTION

I. PURPOSE

- Summaries of data presented in a form that is easy to understand.
- Primarily describe specific characteristics of data
- Find out abnormal observations, outliers and mistakes /errors. Then clean the data before doing further analysis
- Investigate remarkable features of data, using those features to choose suitable model for data analysis

Simple data description methods

A. Describing 1 qualitative variable

Qualitative variable with k values classifies observations in data into k groups

$$K_1, K_2, \dots, K_k.$$

Observations in each group have one same value of the variable
→ Numbers of observations in those groups represent the main feature of the data.

→ Summarizing Categorical Data:

- i) *Frequency distribution table*
- ii) *Percentage/ relative frequency table*
- iii) *Bar chart*
- iv) *Pie chart*

i) Frequency/percentage table

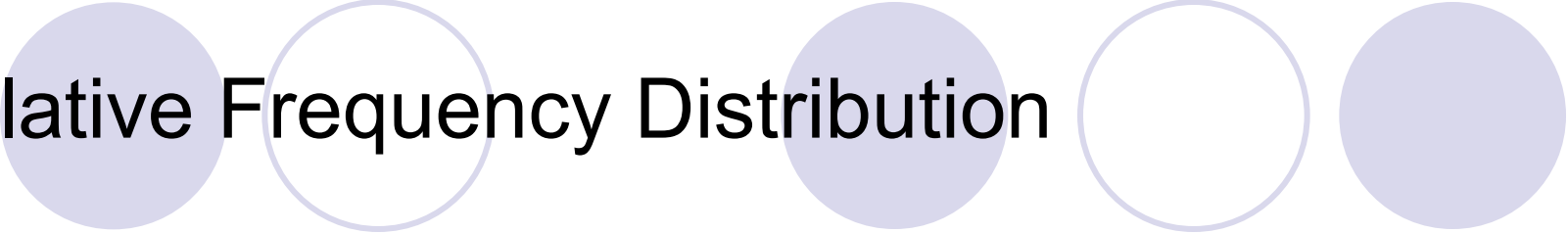
Qualitative variable with k values classifies n observations of a study sample into k groups with $n_1, n_2, \dots, n_k$ observations respectively ($n_1 + n_2 + \dots + n_k = n$). The variable can be represented by a table with k columns:

	Group 1	Group 2		Group k
N	n_1	n_2	$\dots$	n_k
%	$(n_1 / n) *$ 100%	$(n_2 / n) *$ 100%		$(n_k / n) *$ 100%

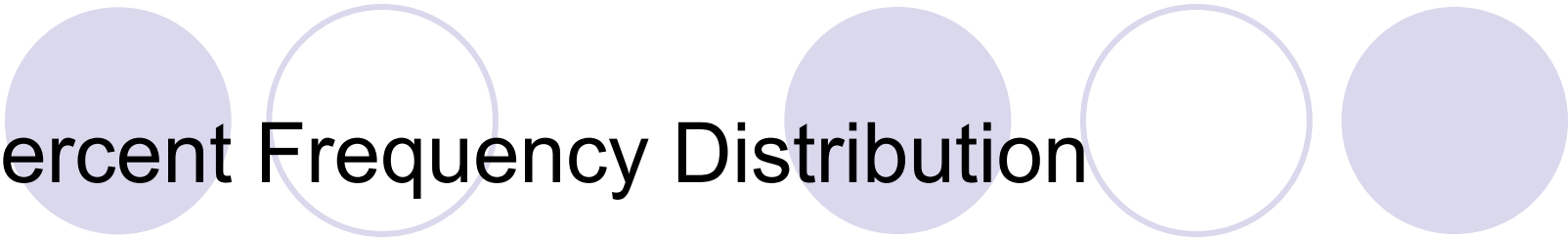
The table gives primary information:

- Frequency (amount of observations) in each group
- Distribution of data: proportion of observations number of each group, ...

Relative Frequency Distribution

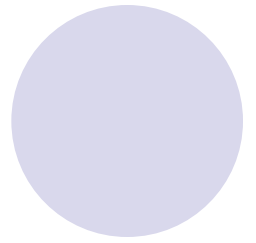
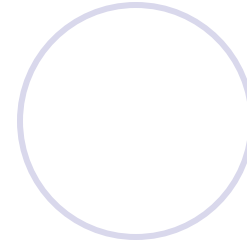
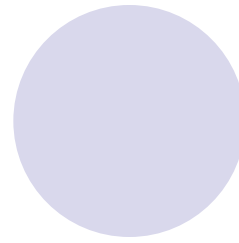
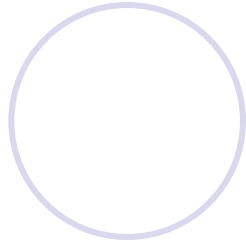
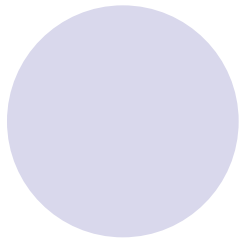


- The relative frequency of a class is the fraction or proportion of the total number of data items belonging to the class.
- A relative frequency distribution is a tabular summary of a set of data showing the relative frequency for each class.



Percent Frequency Distribution

- The percent frequency of a class is the relative frequency multiplied by 100.
- A percent frequency distribution is a tabular summary of a set of data showing the percent frequency for each class.

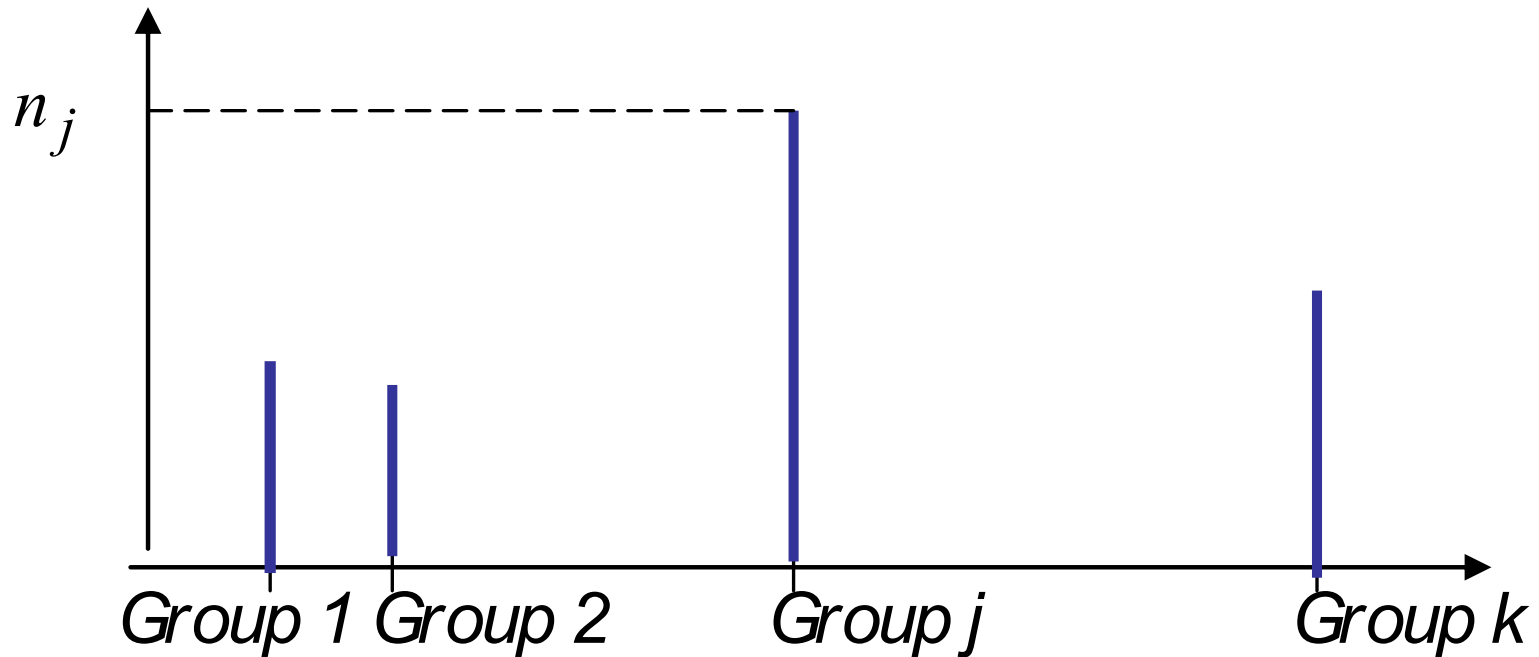


Example 1. To interview question “How often do you go to theater?”, from 148 interviewee, 47 answered “Never”, 71 “Rarely”, 24 “Sometime” and 6 “Frequently”. The data can be presented by frequency table:

	Never	Rarely	Sometime	Frequently	Total
N	47	71	24	6	148
%	31.8 %	48.0 %	16.2 %	4.1 %	100.0 %

ii) Bar chart

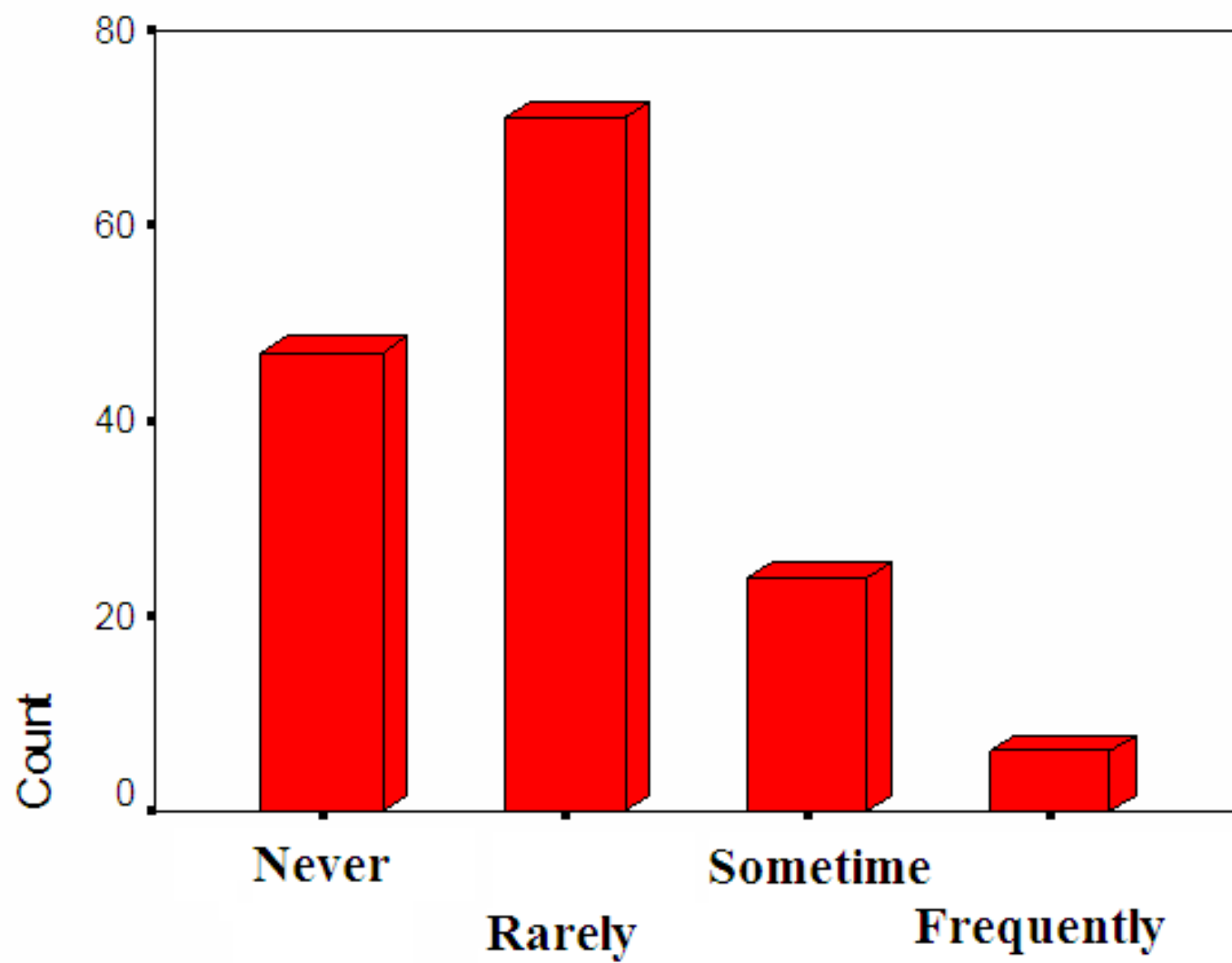
Provides evident picture of qualitative variable distribution:



In the graph, the height of each bar is proportional to observation number of the corresponding group

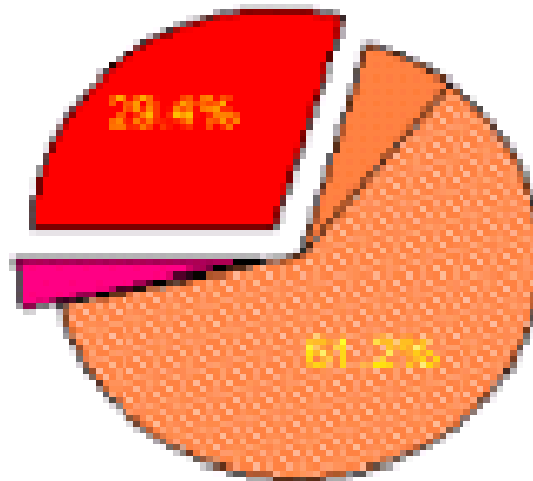
Bar Chart

- A bar chart is a graphical display for depicting qualitative data.
- On one axis (usually the horizontal axis), we specify the labels that are used for each of the classes.
- A frequency, relative frequency, or percent frequency scale can be used for the other axis (usually the vertical axis).
- Using a bar of fixed width drawn above each class label, we extend the height appropriately.
- The bars are separated to emphasize the fact that each class is a separate category.



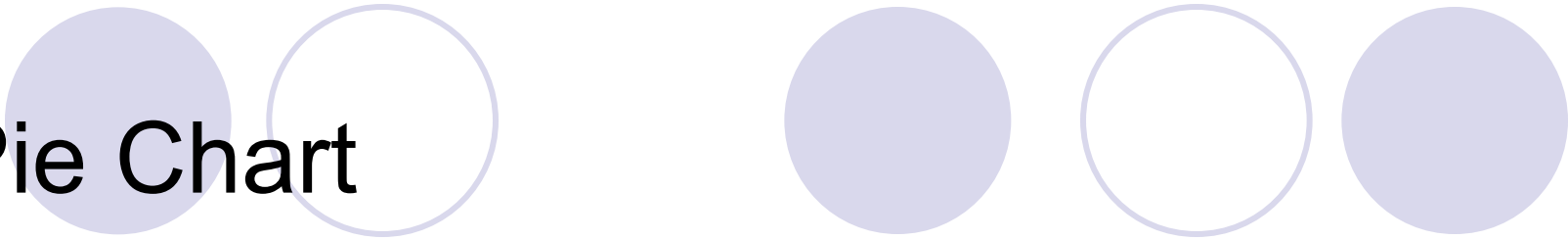
iii) Pie chart

Presents proportions (percentages) of observations numbers of groups in total number of all observations in the sample



Area of each part in the chart is proportional to the observations number of corresponding group.

Pie Chart

The title 'Pie Chart' is positioned on the left side of the slide. To its right, there are two groups of three circles each. The first group consists of a solid light purple circle, a circle with a light purple outline, and another solid light purple circle. The second group consists of a solid light purple circle, a circle with a light purple outline, and another solid light purple circle.

- The pie chart is a commonly used graphical display for presenting relative frequency and percent frequency distributions for categorical data.
- First draw a circle; then use the relative frequencies to subdivide the circle into sectors that correspond to the relative frequency for each class.

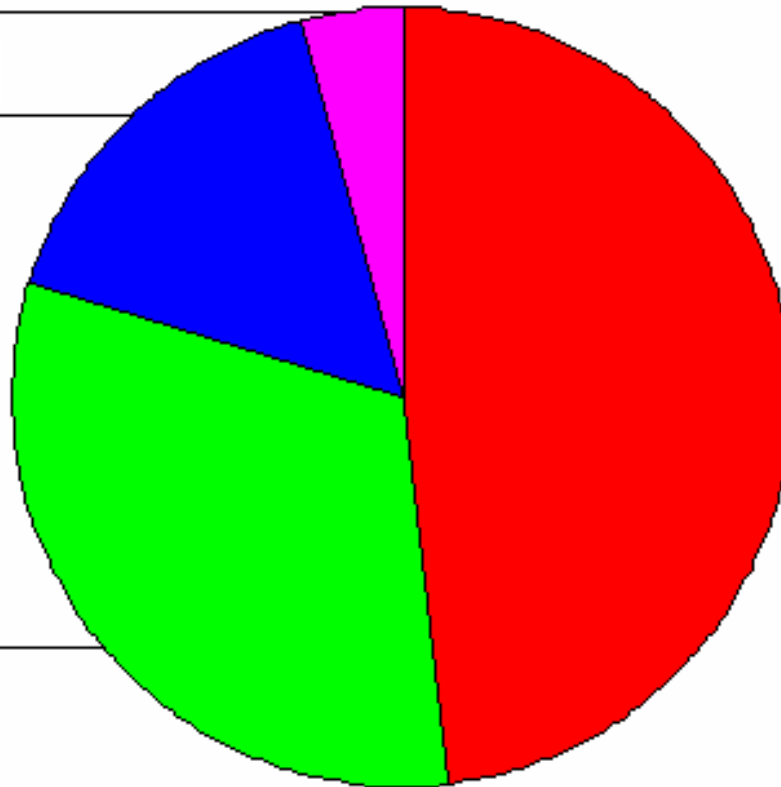
Bar chart represents the data in Example 1

Frequently

Sometime

Never

Rarely



Using Excel to Describe Categorical Variables

- Excel has functions for computing the parameters of categorical variables and draw graphs:
 - SUM, COUNT, COUNTIF used to compute the frequencies.
 - BAR, PIE used to draw the bar charts and pie charts.

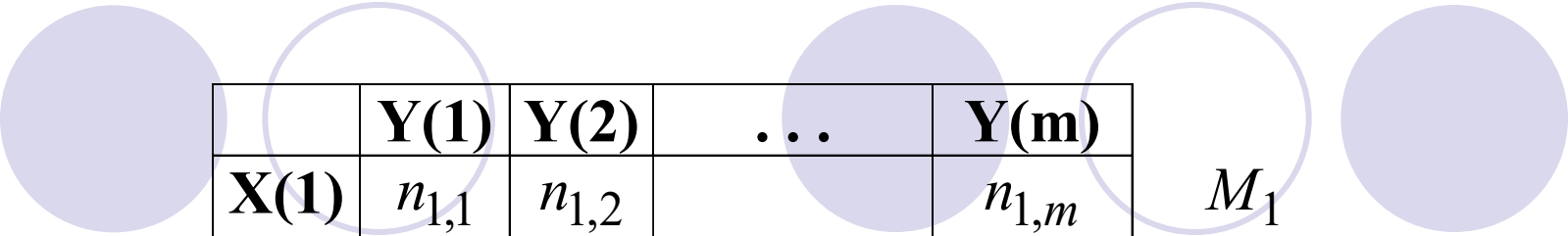
B. Describe relation between 2 qualitative variables

Cross table with levels of one variable in rows, levels of the second variable in columns:

	Y(1)	Y(2)	...	Y(m)	
X(1)	$n_{1,1}$	$n_{1,2}$		$n_{1,m}$	M_1
X(2)	$n_{2,1}$	$n_{2,2}$		$n_{2,m}$	M_2
			...		
X(k)	$n_{k,1}$	$n_{k,2}$		$n_{k,m}$	M_k
	K_1	K_2		K_m	N

Usually,

- The first variable (rows) is a independent (describing, cause, input) variable*
- The second variable (columns) is a dependent (descriptive, result, output) variable*



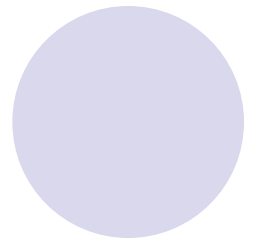
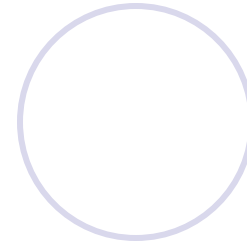
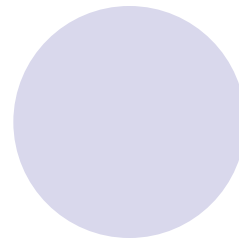
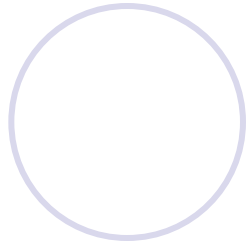
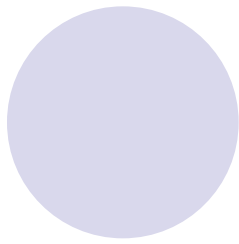
	Y(1)	Y(2)	...	Y(m)	
X(1)	$n_{1,1}$	$n_{1,2}$		$n_{1,m}$	M_1
X(2)	$n_{2,1}$	$n_{2,2}$		$n_{2,m}$	M_2
			...		
X(k)	$n_{k,1}$	$n_{k,2}$		$n_{k,m}$	M_k
	K_1	K_2		K_m	N

In cell ij of table, $n_{i,j}$: number of observations belonging simultaneously to the level i of the first variable and to the level j of the second variable,

M_i : sum of all numbers in the row i (number of observations in i -th level of the first variable),

K_j : sum of all numbers in the column j (number of observations in j -th level of the second variable)

N : total number of all observations in the sample



Crosstabulation: Row or Column Percentages

- Converting the entries in the table into row percentages or column percentages can provide additional insight about the relationship between the two variables.



Table with percentages % across rows: $n_{i,j} / M_i$ gives information about distribution of “output” variable Y in each level of “input” variable X

	Y(1)	Y(2)	...	Y(m)
X(1)	$n_{1,1} / M_1$	$n_{1,2} / M_1$		$n_{1,m} / M_1$
X(2)	$n_{2,1} / M_2$	$n_{2,2} / M_2$		$n_{2,m} / M_2$
			...	
X(k)	$n_{k,1} / M_k$	$n_{k,2} / M_k$		$n_{k,m} / M_k$



Table with percentages % across columns: $n_{i,j} / K_j$ gives information about distribution of “input” variable X in each level of “output” variable Y

	Y(1)	Y(2)	...	Y(m)
X(1)	$n_{1,1} / K_1$	$n_{1,2} / K_2$		$n_{1,m} / K_m$
X(2)	$n_{2,1} / K_1$	$n_{2,2} / K_2$		$n_{2,m} / K_m$
			...	
X(k)	$n_{k,1} / K_1$	$n_{k,2} / K_2$		$n_{k,m} / K_m$



Table with percentages % in whole sample: $n_{i,j} / N$ gives information about total distribution in sample

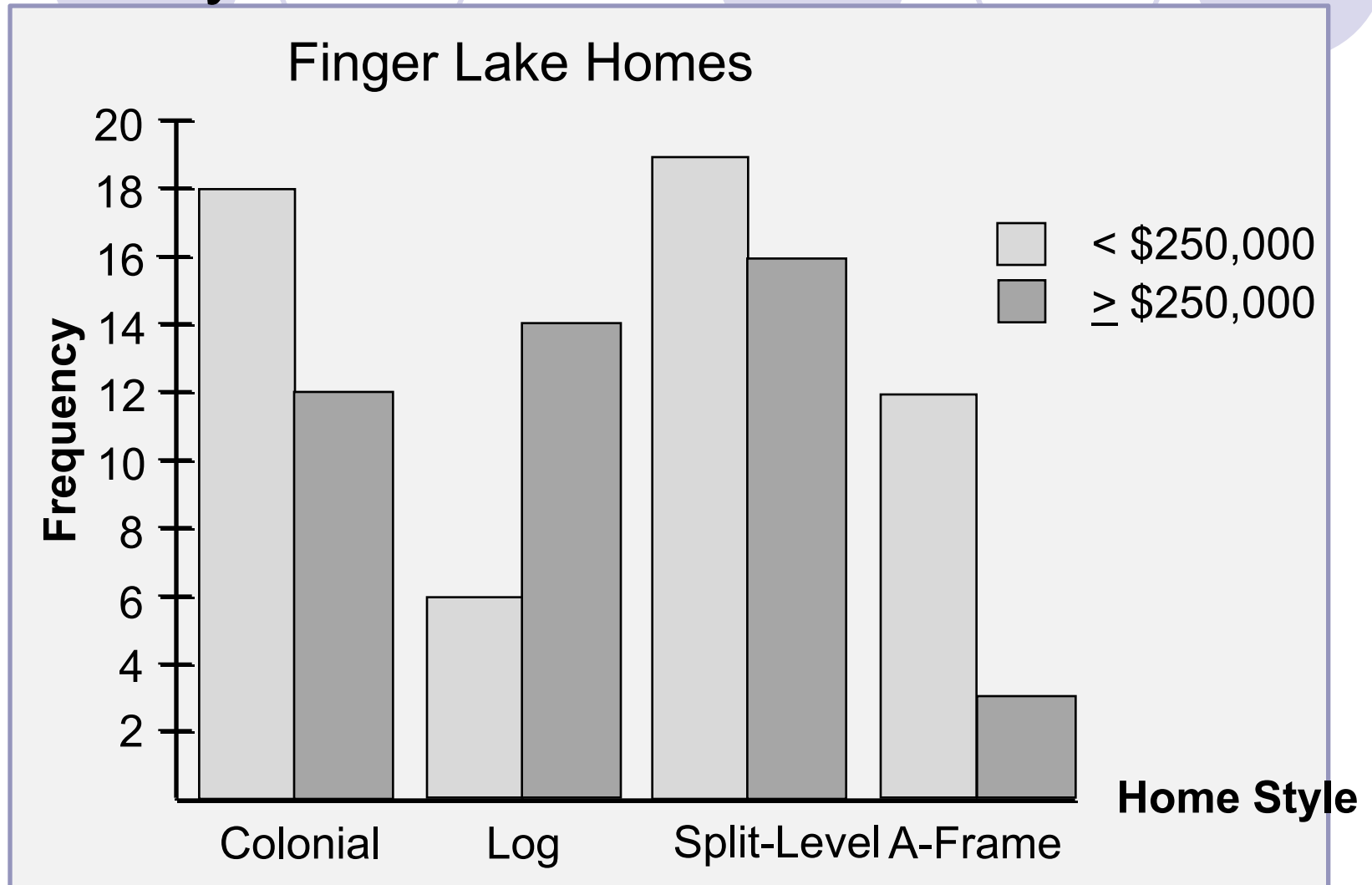
	Y(1)	Y(2)	...	Y(m)
X(1)	$n_{1,1} / N$	$n_{1,2} / N$		$n_{1,m} / N$
X(2)	$n_{2,1} / N$	$n_{2,2} / N$		$n_{2,m} / N$
			...	
X(k)	$n_{k,1} / N$	$n_{k,2} / N$		$n_{k,m} / N$

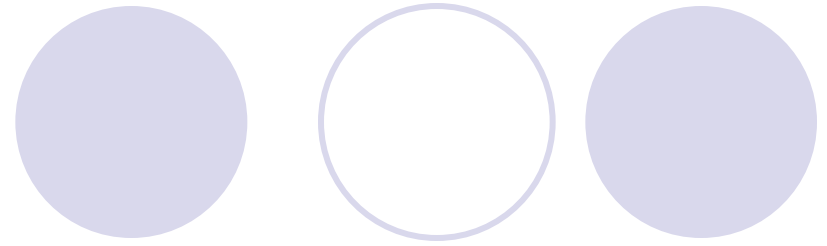
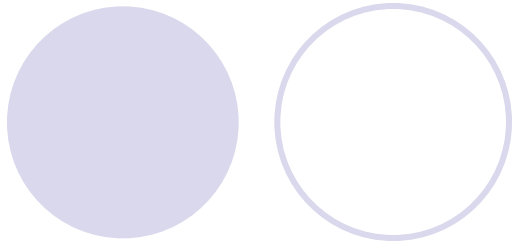


Side-by-Side Bar Chart

- A side-by-side bar chart is a graphical display for depicting multiple bar charts on the same display.
- Each cluster of bars represents one value of the first variable.
- Each bar within a cluster represents one value of the second variable.

Side-by-Side Bar Chart

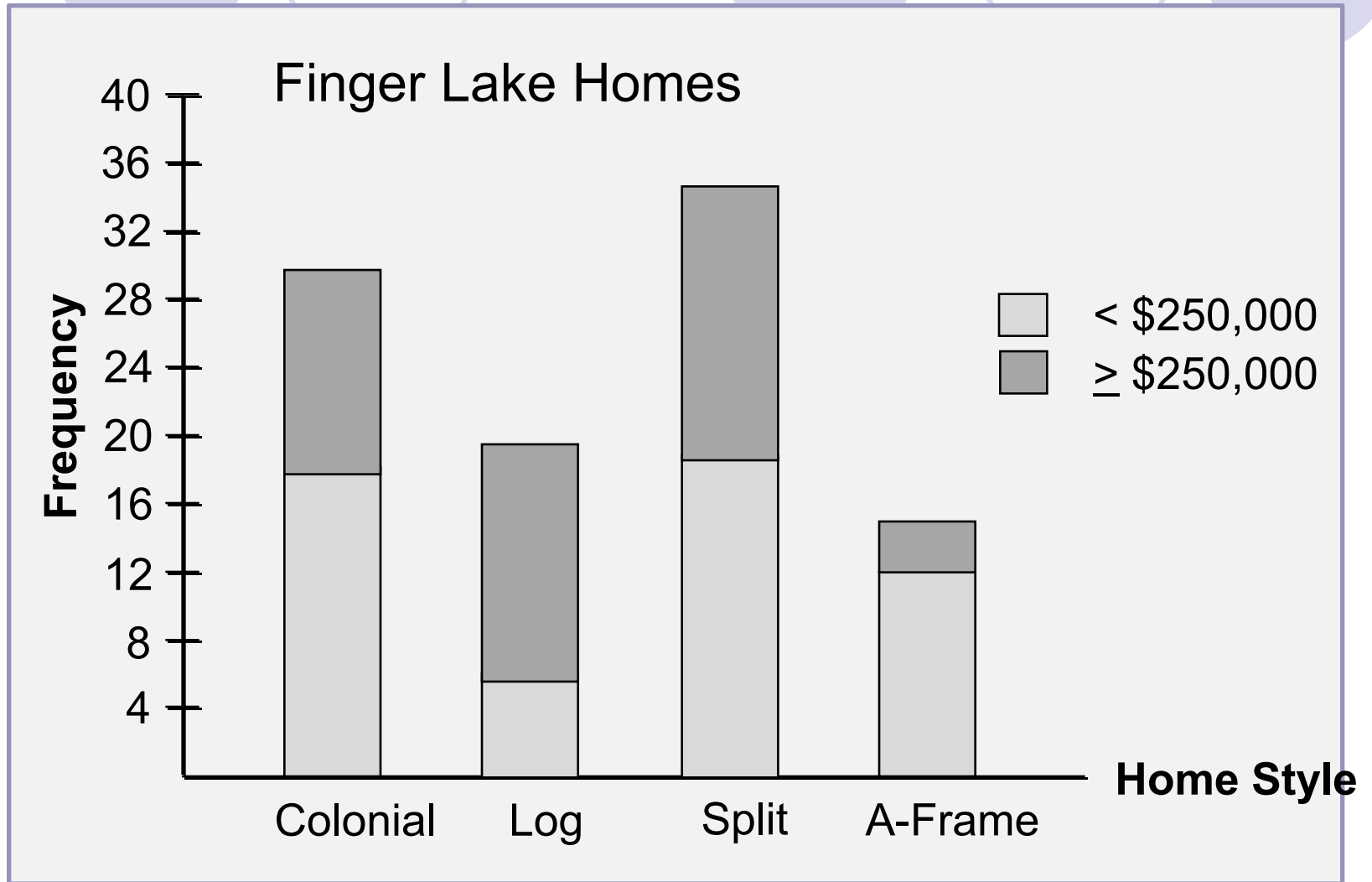




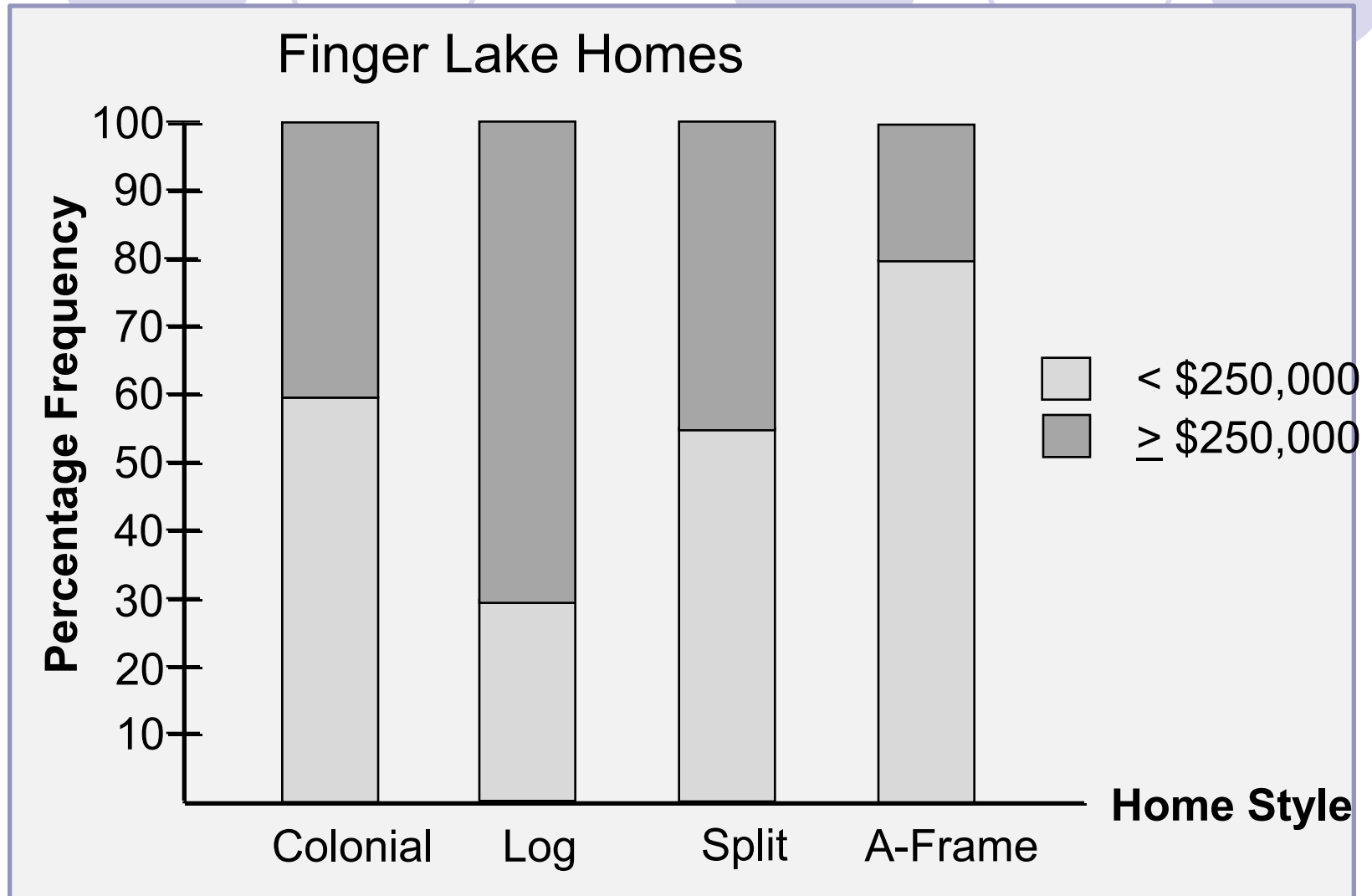
Stacked Bar Chart

- A stacked bar chart is another way to display and compare two variables on the same display.
- It is a bar chart in which each bar is broken into rectangular segments of a different color.
- If percentage frequencies are displayed, all bars will be of the same height (or length), extending to the 100% mark.

Stacked Bar Chart



Stacked Bar Chart





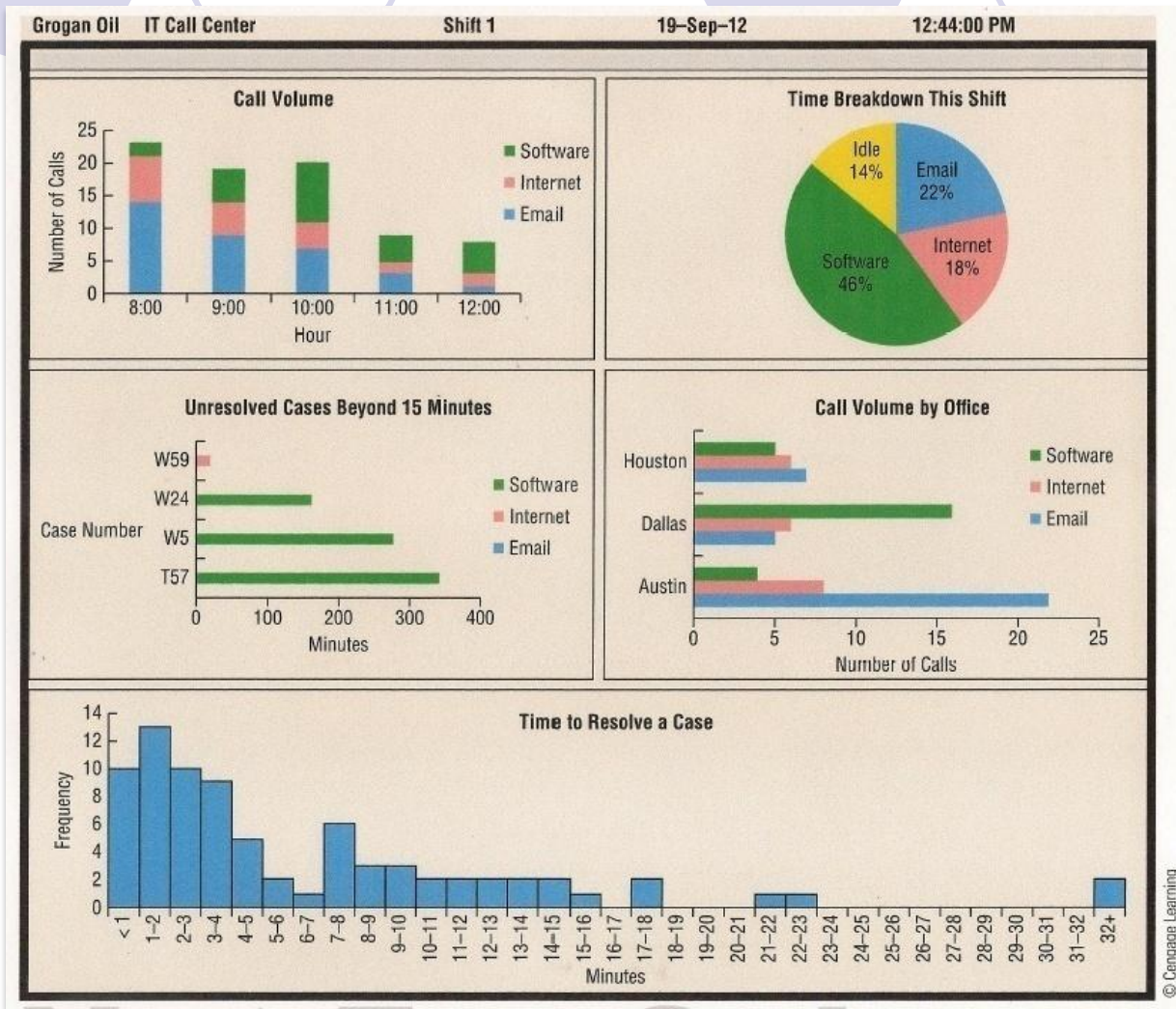
Choosing the Type of Graphical Display

- Displays used to make comparisons:

Side-by-Side Bar Chart to compare two variables

Stacked Bar Chart to compare the relative frequency or Percent frequency of two categorical variables

Example



Using Excel to Describe Categorical Variables

- Excel has functions for computing the parameters of categorical variables and draw graphs:
 - SUM, COUNT, COUNTIF, COUNTIFS used to compute the frequencies.
 - BAR used to draw the bar charts.

C. Describing a quantitative variable

For a quantitative variable X with the sample of n observations

$$\mathbf{X} = \{x_1, x_2, \dots, x_n\},$$

where x_i is the value of X at observation i . Then several methods can be used to describe the variable:

i) Extreme values of variable

ii) Parameters measuring central tendency of data

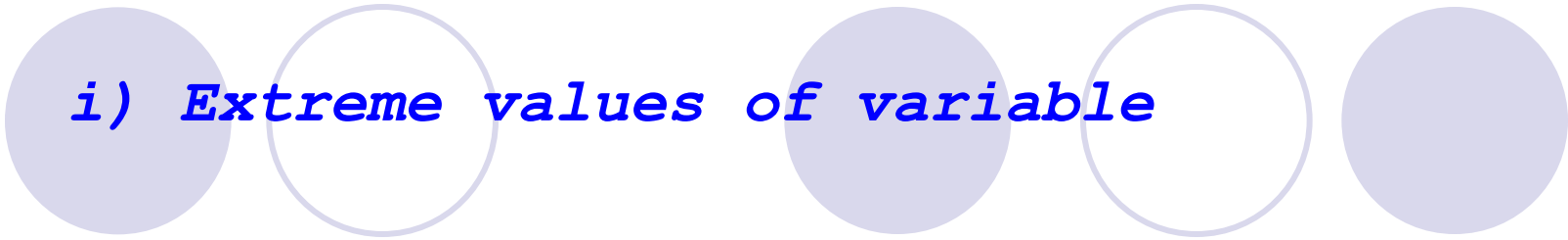
iii) Parameters measuring variability of data

iv) Histogram

v) Quantiles

vi) Stem-leaf plot

vii) Box plot



i) Extreme values of variable

$\text{Max}(\mathbf{X})$ - the largest value of data,

$\text{min}(\mathbf{X})$ - the smallest value of data

Knowing the largest and the smallest values of data one can have some conclusions, i.g.

- The data values are contained in a reasonable interval or not?
- If there is some thing implying meaningless of the data?
- etc.



ii) Parameters measuring "central" tendency of data

1. Mean value of variable

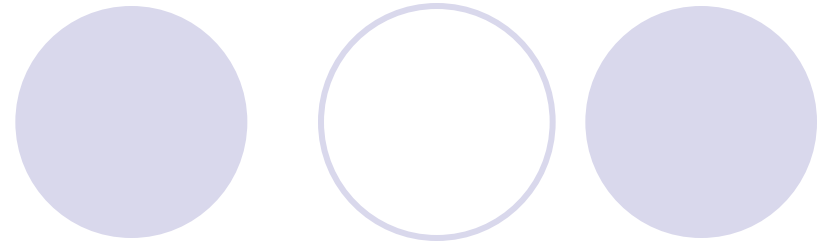
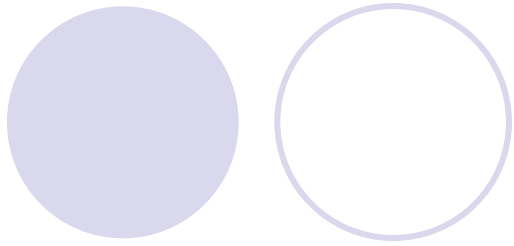
$$\text{Mean}(\mathbf{X}) = \frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{n} (x_1 + x_2 + \dots + x_n) ,$$

2. Average number of two extreme values

$$\text{ME}(\mathbf{X}) = \{ \min(\mathbf{X}) + \text{Max}(\mathbf{X}) \} / 2$$

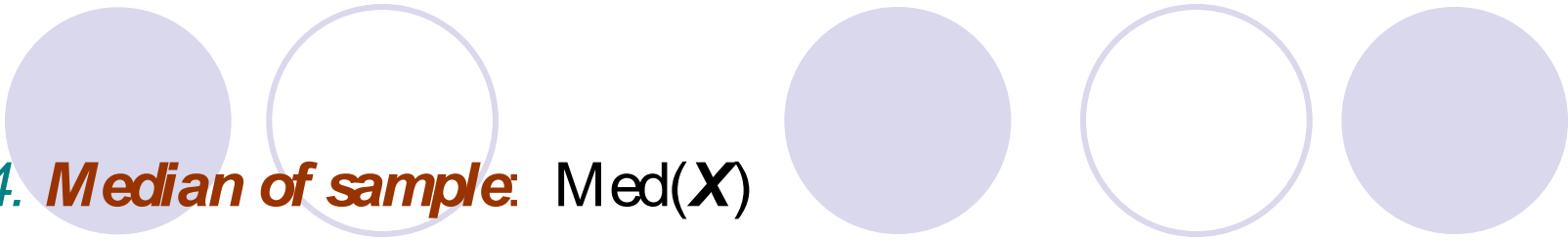
3. Mode of sample: Mode ($\mathbf{X}$)

A data value whose frequency is higher than frequency of any neighbourhood value of data



Mode

- The mode of a data set is the value that occurs with greatest frequency.
- The greatest frequency can occur at two or more different values.
- If the data have exactly two modes, the data are bimodal.
- If the data have more than two modes, the data are multimodal.



4. *Median of sample:* $\text{Med}(\mathbf{X})$

Value whose cumulative frequency equals (approximately) 50%: the point of value dividing the sample into two “equal” parts, 1/2 lying in the left and 1/2 lying in the right hand side of this point.

If n elements of data are arranged in order:

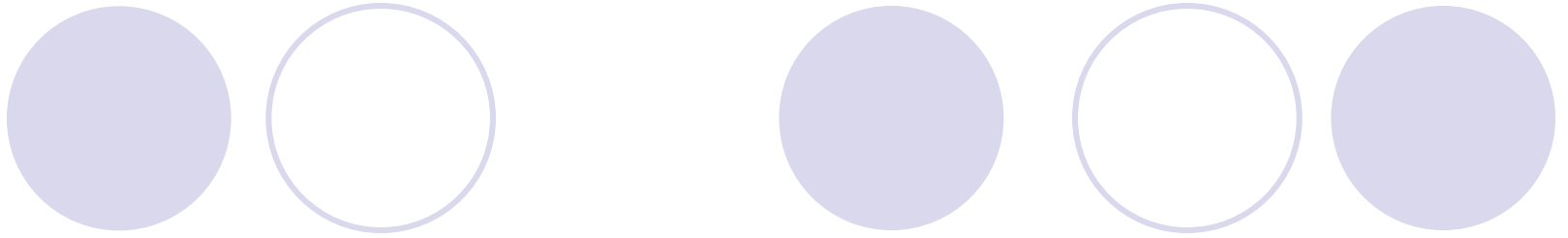
$$x_1 \leq x_2 \leq \dots \leq x_n \quad .$$

Then $\text{Med}(\mathbf{X}) = x_{(n+1)/2}$ if n is odd,

and

$$\text{Med}(\mathbf{X}) = x_{n/2} \quad \text{if } n \text{ is even}$$

$$[\text{Med}(\mathbf{X}) = [x_{n/2} + x_{n/2+1}] / 2]$$



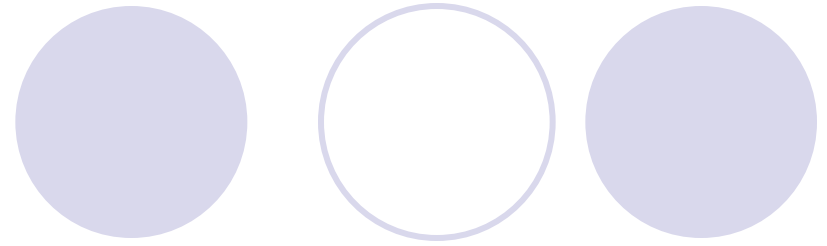
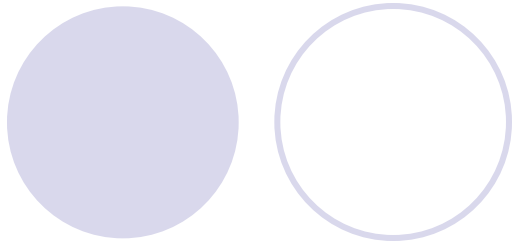
Example:

$$\text{Med}(\{1, 2, 5\}) = 2,$$

$$\text{Med}(\{1, 3, 3, 3\}) = 3,$$

$$\text{Med}(\{1, 2, 5, 7\}) = 2$$

$$[\text{Med}(\{1, 2, 5, 7\}) = 3.5]$$



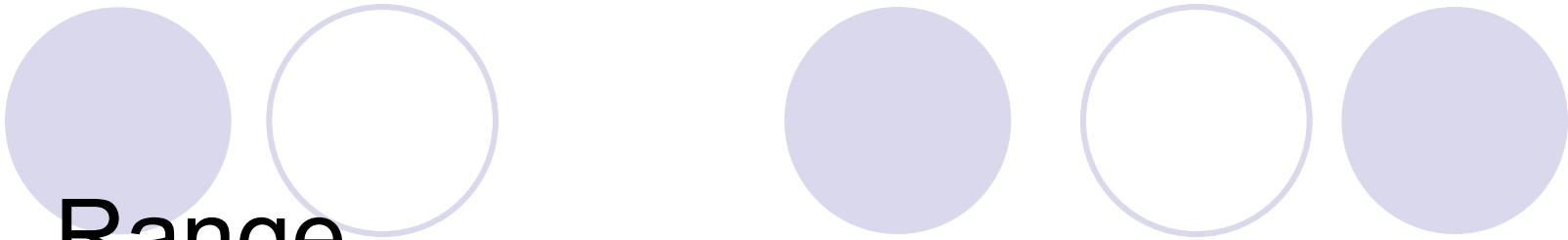
Median

- The median of a data set is the value in the middle when the data items are arranged in ascending order.
- Whenever a data set has outlier extreme values, the median is the preferred measure of central location.
- The median is the measure of location most often reported for annual income and property value data.
- A few extremely large incomes or property values can inflate the mean.



iii) Parameters measuring variability of data

- 1. Range*
- 2. Interquartile range*
- 3. Variance*
- 4. Standard deviation*

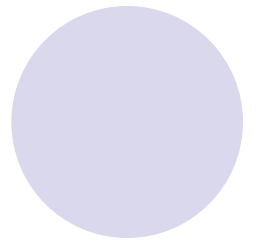
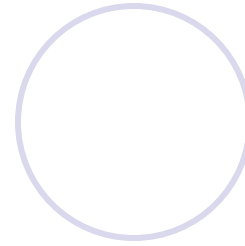
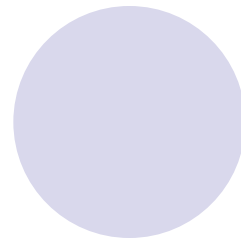
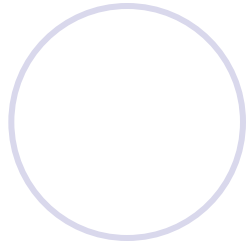
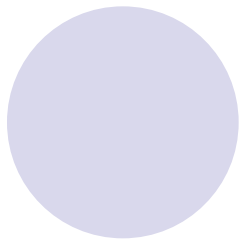


1. Range

- The range of a data set is the difference between the largest and smallest data values.

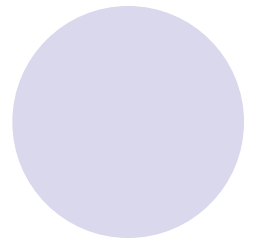
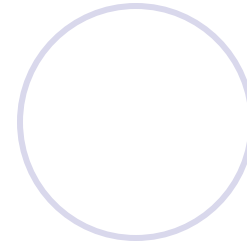
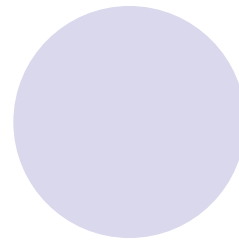
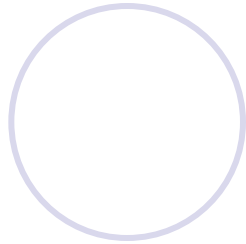
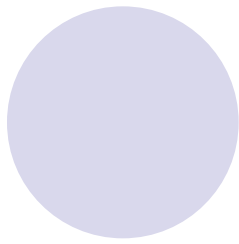
$$\text{Range} = \text{Largest value} - \text{Smallest value}$$

- It is the simplest measure of variability.
- It is very sensitive to the smallest and largest data values.



2. Interquartile Range

- The interquartile range of a data set is the difference between the third quartile and the first quartile.
- It is the range for the middle 50% of the data.
- It overcomes the sensitivity to extreme data values.



3. *Variance:*

Population variance

$$PVar(\mathbf{X}) = \frac{1}{n} \sum_{k=1}^n (x_k - Mean(X))^2$$

Sample variance

$$Var(\mathbf{X}) = \frac{1}{n-1} \sum_{k=1}^n (x_k - Mean(X))^2$$

4. *Standard Deviation:*

$$SD(\mathbf{X}) = \sqrt{Var(X)}$$

$$PSD(\mathbf{X}) = \sqrt{PVar(X)}$$

Variance

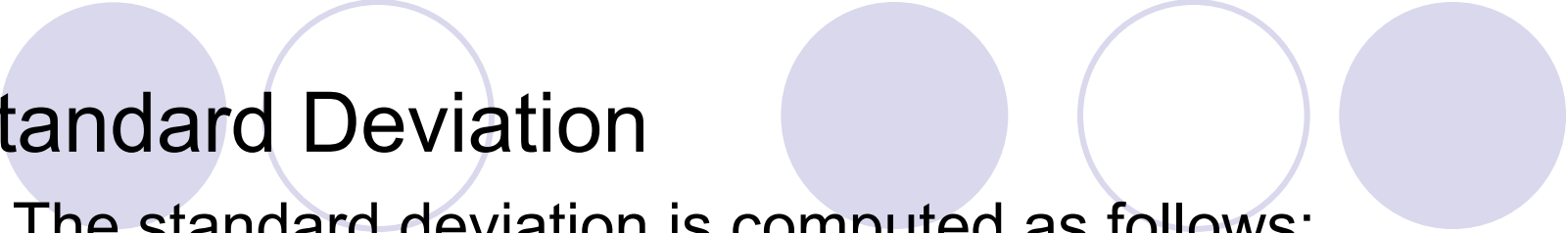
- The variance is the average of the squared differences between each data value and the mean.

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

for a
sample

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

for a
population



Standard Deviation

- The standard deviation is computed as follows:

$$s = \sqrt{s^2}$$

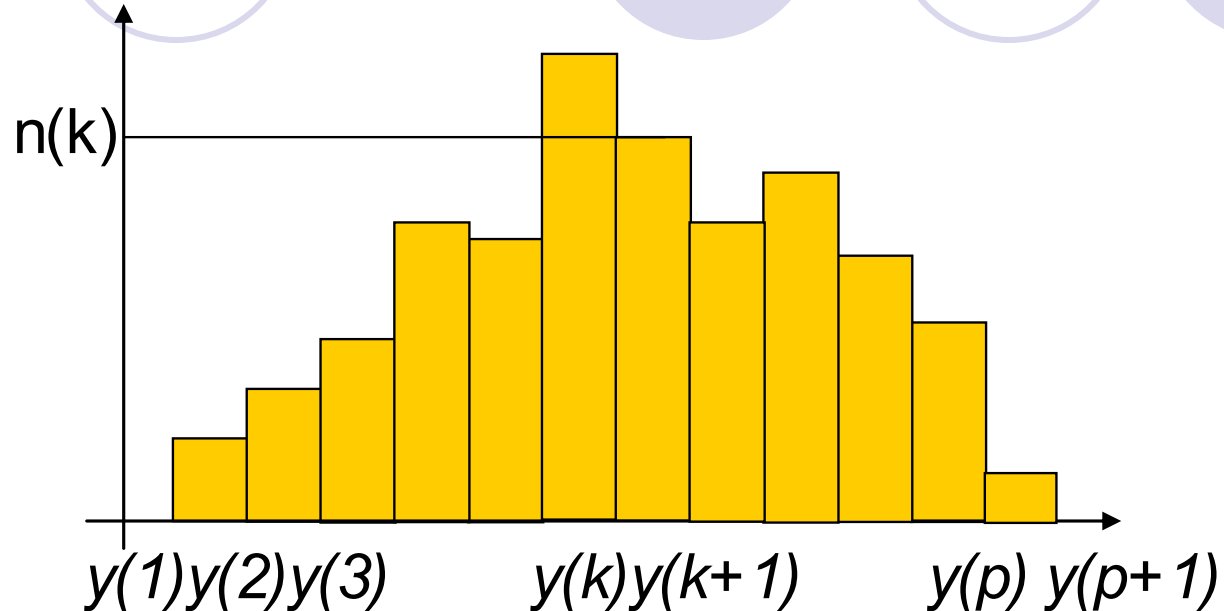
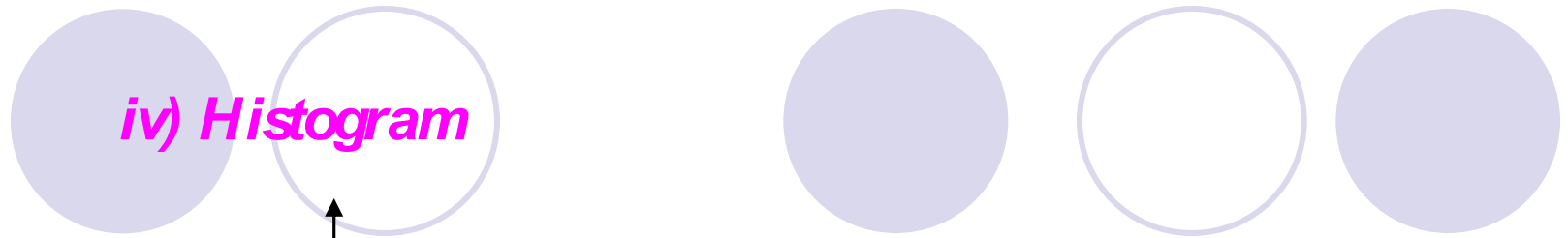
for a
sample

$$\sigma = \sqrt{\sigma^2}$$

for a
population

Standard Deviation

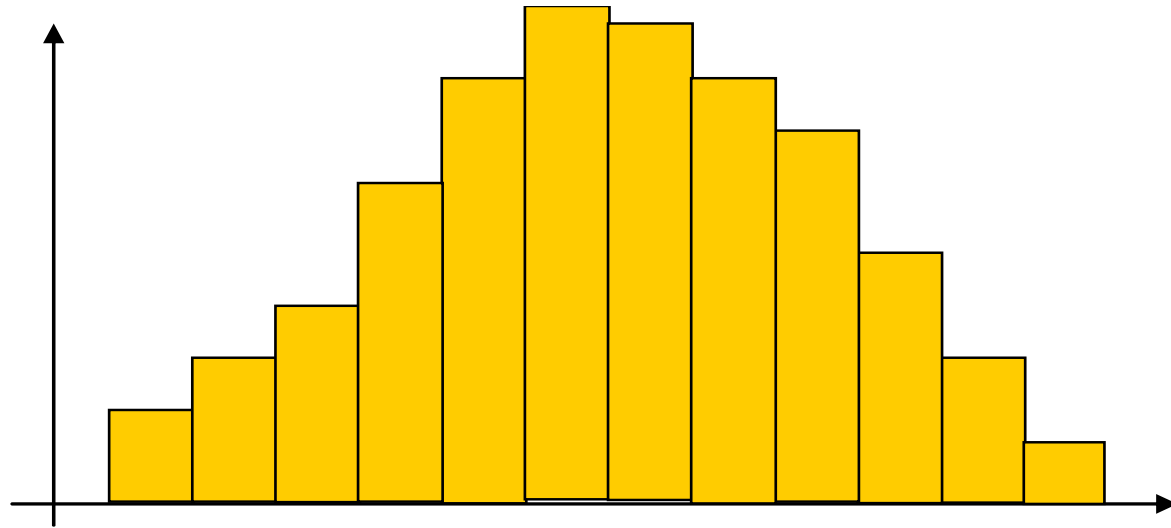
- The standard deviation of a data set is the positive square root of the variance.
- It is measured in the same units as the data, making it more easily interpreted than the variance.



- + Let $y() = \min(X)$, $y(p+1) = \max(X)$ and set $A = [y(1), y(p+1))$
- + Divide A into p equal intervals
- + Determine $n(k)$ as frequency of values of X belonging to the k -th interval
- + The height of k -th rectangle is taken proportionally to $n(k)$

Histogram types

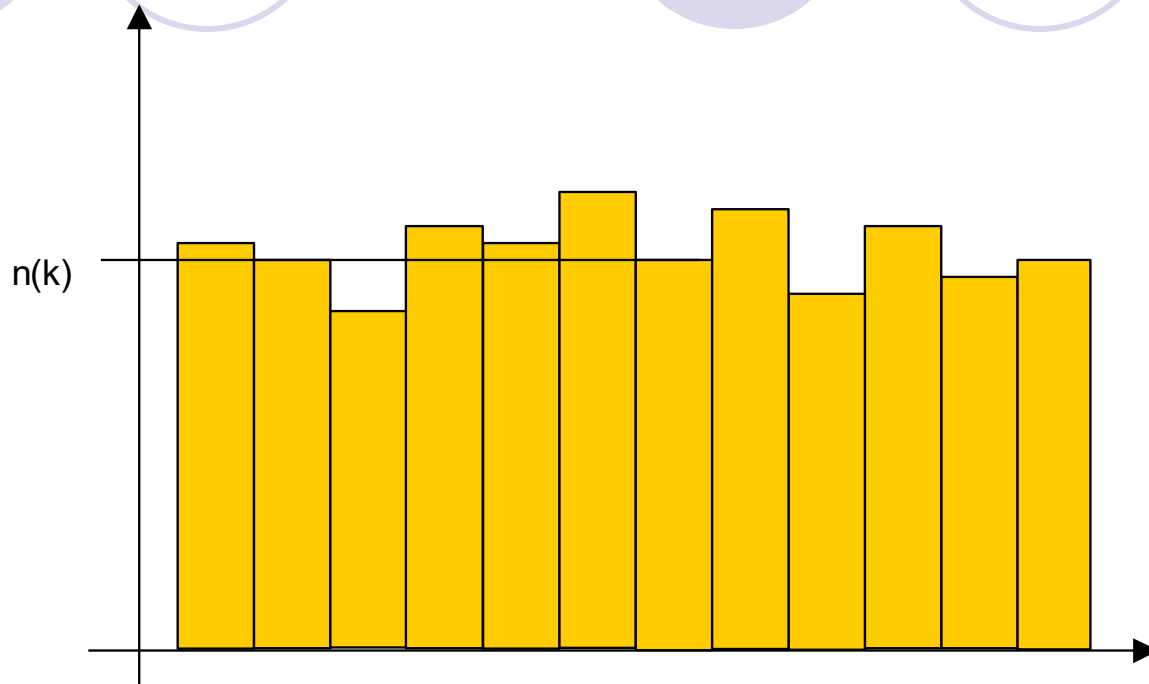
(1) *Symmetric unimodal histogram*



Properties:

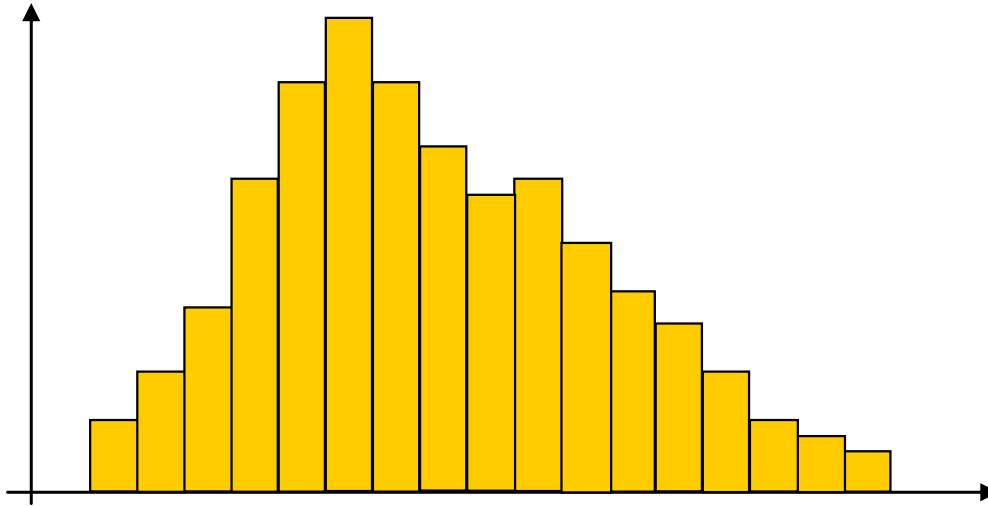
- Mode, mean and median values are close each to another
- The sample can be represented by two parameters: mean value $\text{Mean}(\mathbf{X})$ and standard deviation $\sigma(\mathbf{X})$.

(2) Uniform histogram



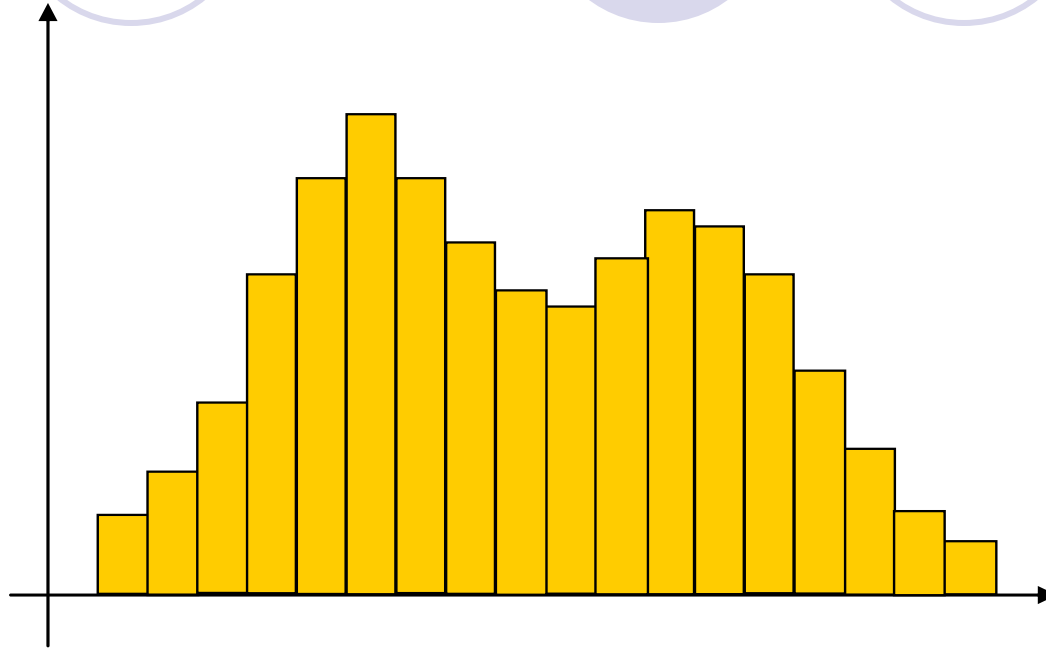
All rectangles have almost the same height. Then the sample can be resumed by values of $\min(X)$, $\max(X)$ and the range $w(X)$

(3) *Asymmetric unimodal histogram*



- Mode, median and mean values are different. The sample can not be resummed by mean value and standard deviation
- Use some transformation for X (i.g. $\log(X)$) to make (if possible) a variable with symmetric form

(4) *Bi- or multimodal histogram*

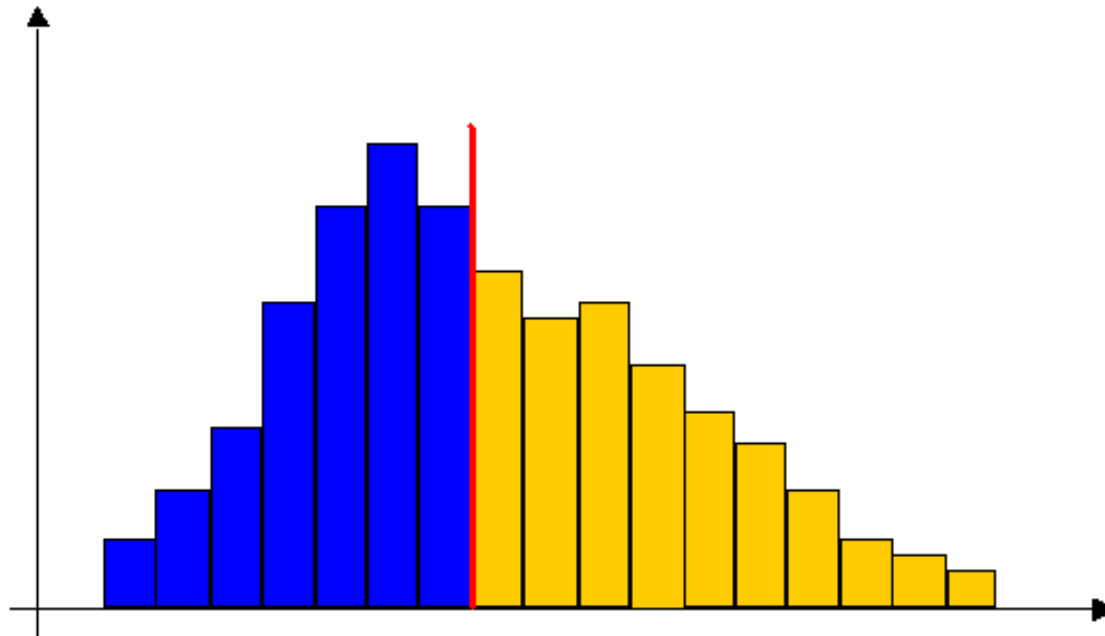


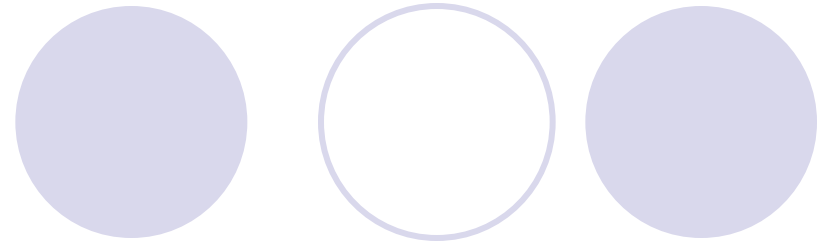
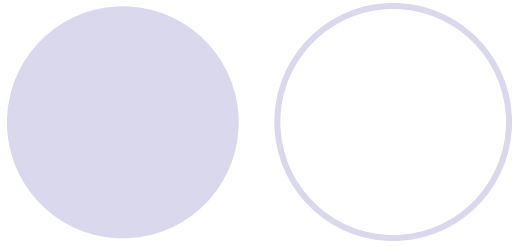
With multi-modal histogram, the data should be non-homogenous, may be a compound of several subpopulations
→ Separate the sample to two or many smaller sub-samples to study separately

v) Quantile

a% Quantile: - point dividing sample units into two parts: the left part contains **a%** amount of all observations in sample (then the right part contains **(100-a)%** amount of observations)

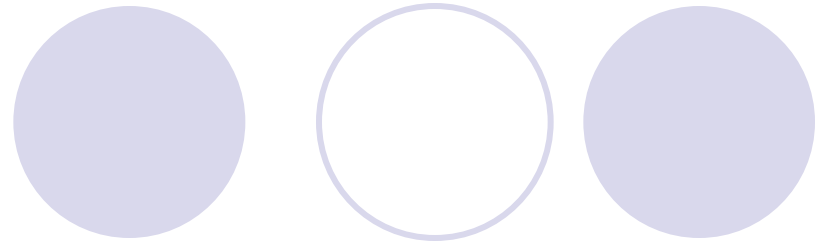
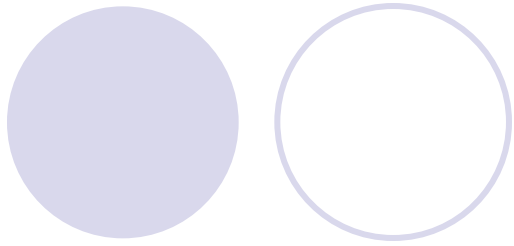
Median = 50% quantile, dividing the sample to 2 equal parts, each contains 1/2 amount of sample units





Percentiles

- The p th percentile of a data set is a value such that at least p percent of the items take on this value or less and at least $(100 - p)$ percent of the items take on this value or more.
- A percentile provides information about how the data are spread over the interval from the smallest value to the largest value.



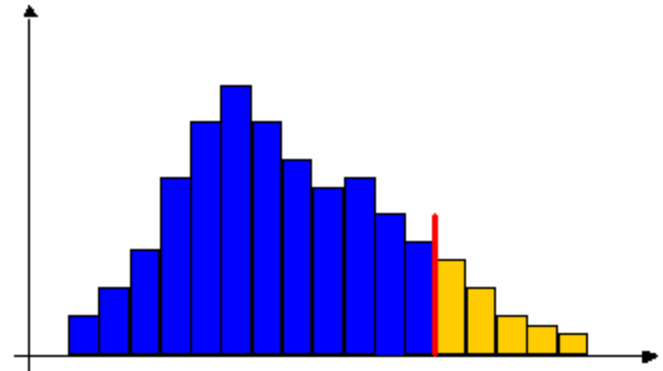
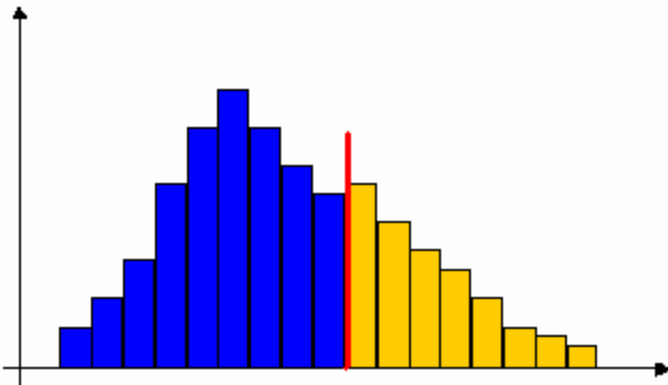
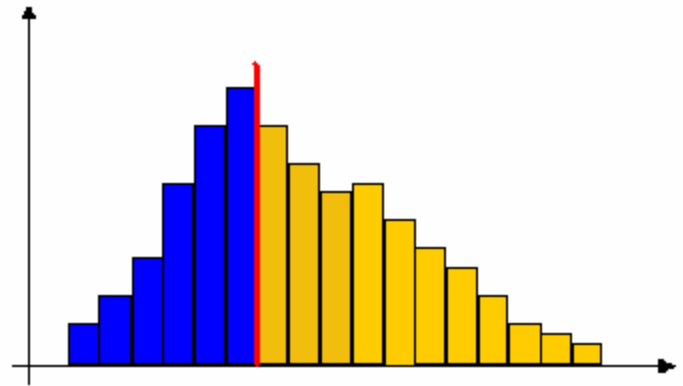
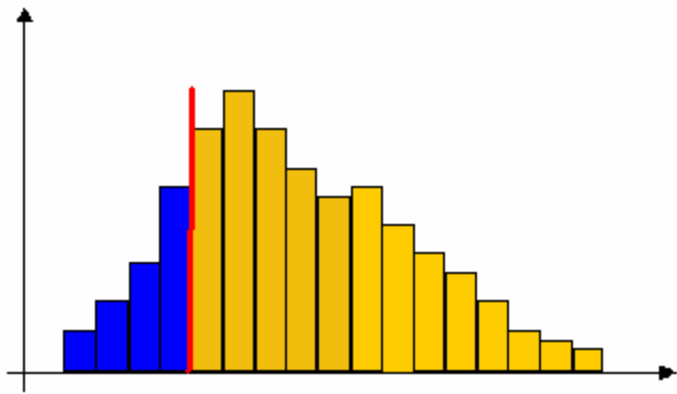
Percentiles

- Arrange the data in ascending order.
- Compute L_p , the location of the p th percentile.

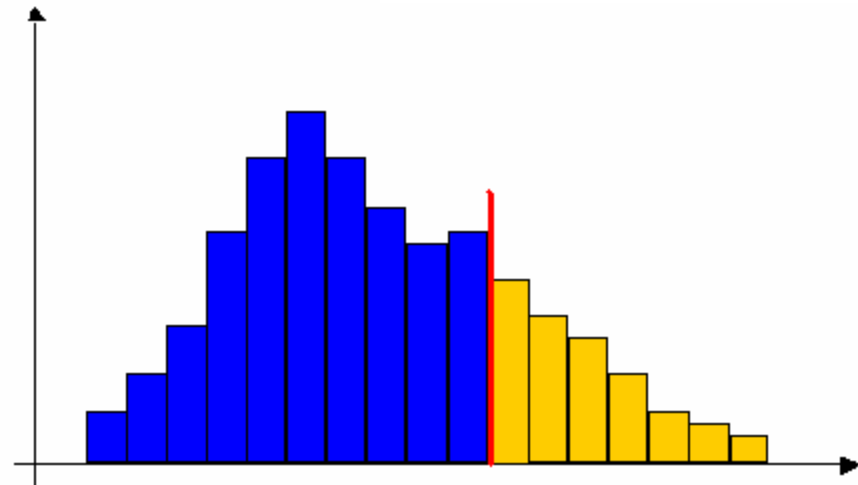
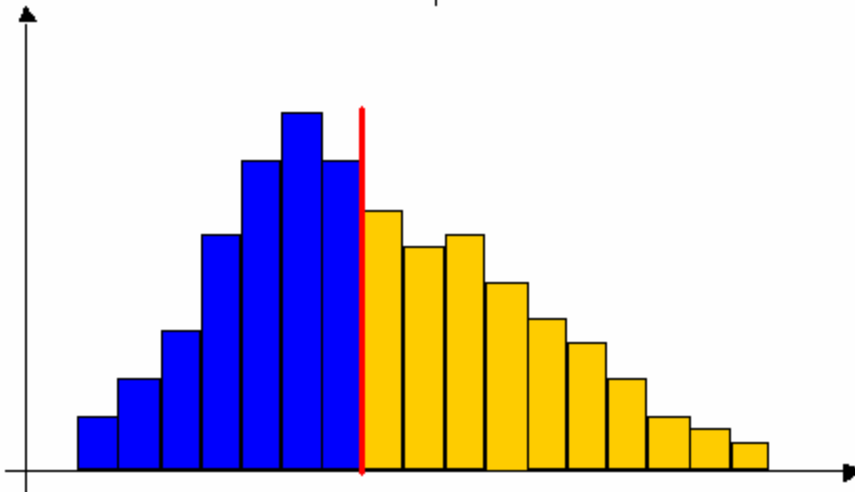
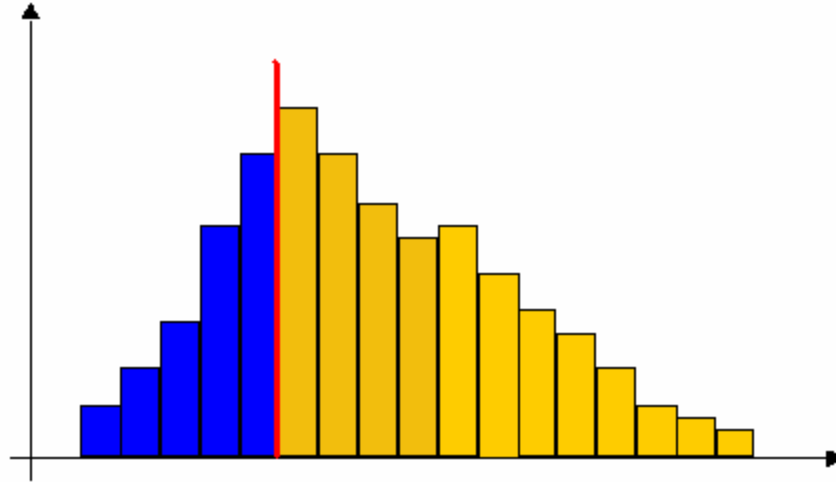
$$L_p = [(p/100)(n + 1)]$$

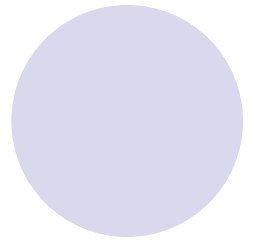
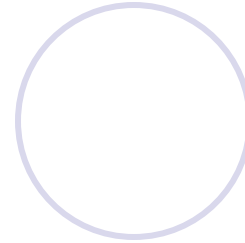
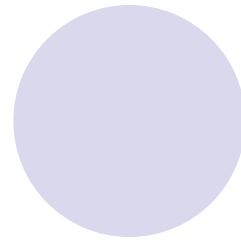
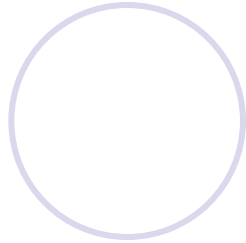
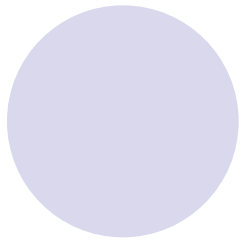
Special cases

Quintiles: percentiles 20%, 40%, 60% and 80%,
dividing the sample into 5 equal parts



Quartiles: percentiles 25%, 50% and 75%, divide the sample into 4 equal parts

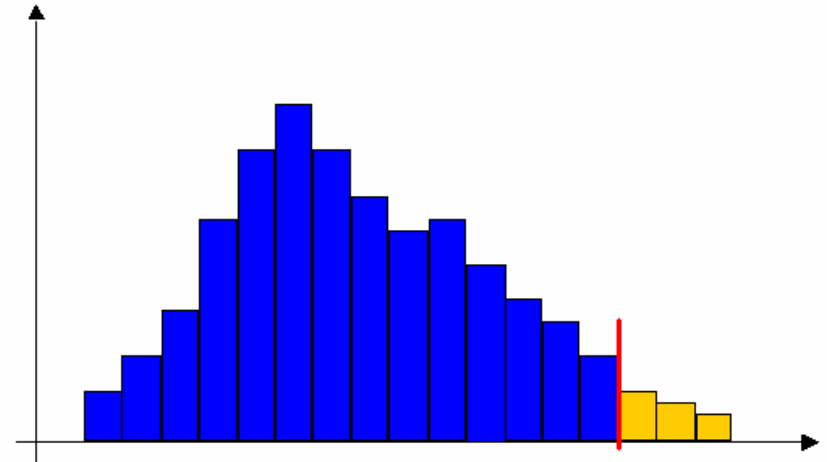
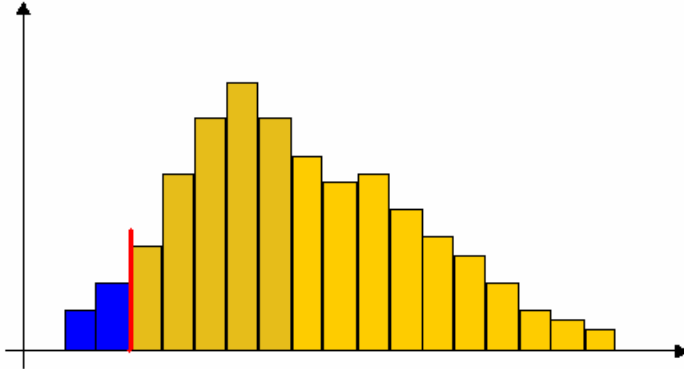




Quartiles

- Quartiles are specific percentiles.
- First Quartile = 25th Percentile
- Second Quartile = 50th Percentile = Median
- Third Quartile = 75th Percentile

5% and 95% percentiles

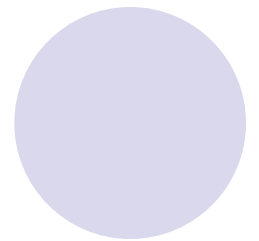
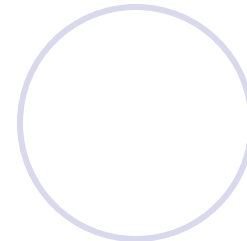
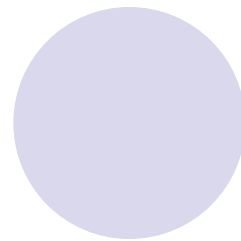
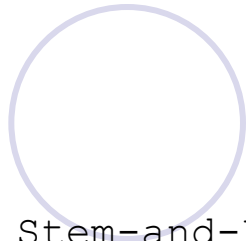
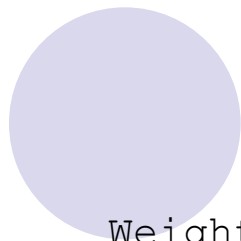


vi) Stem-leaf Plot

Example: Weight of children in Uong Bi hospital

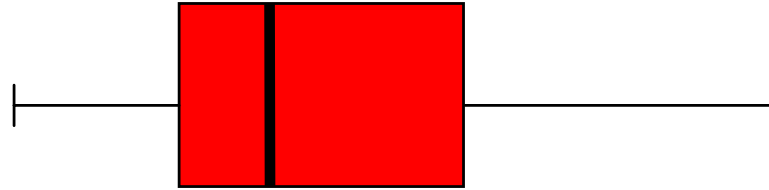
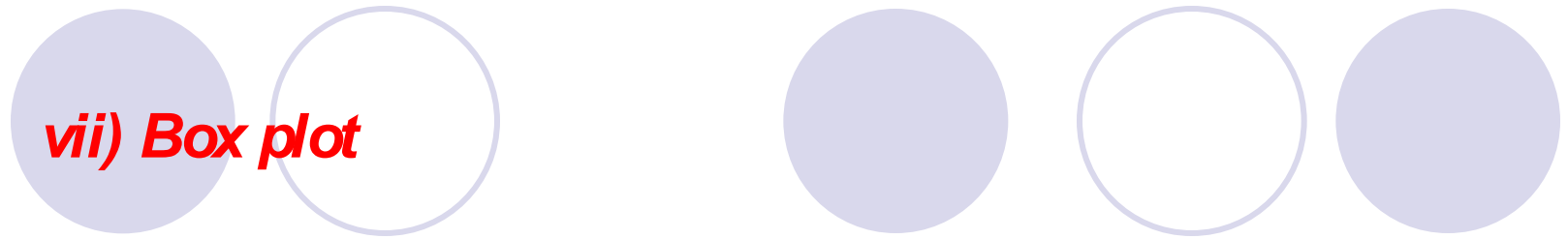
Weight Stem-and-Leaf Plot

Frequency	Stem &	Leaf
3.00	9 .	005
3.00	10 .	003
6.00	11 .	005578
10.00	12 .	0000005688
12.00	13 .	000000000001
14.00	14 .	00000035557889
14.00	15 .	00000000004559
17.00	16 .	00000000445555567
18.00	17 .	000000000000245559
21.00	18 .	000000000000255556678
18.00	19 .	000000000055555778
33.00	20 .	00000000000000000002233345566799
21.00	21 .	00000000000005555555
28.00	22 .	0000000000000000002555566778
25.00	23 .	000000000000000000002459
21.00	24 .	0000000000000000002448
12.00	25 .	000000000000



Weight Stem-and-Leaf Plot

Frequency	Stem &	Leaf	
3.00	9 .	005	3
3.00	10 .	003	6
6.00	11 .	005578	12
10.00	12 .	0000005688	22
12.00	13 .	0000000000001	34
14.00	14 .	00000035557889	48
14.00	15 .	000000000004559	62
17.00	16 .	00000000445555567	79
18.00	17 .	0000000000000245559	97
21.00	18 .	0000000000000255556678	118
18.00	19 .	0000000000055555778	136
33.00	20 .	000000000000000000002233345566799	140
21.00	21 .	0000000000000055555555	107
28.00	22 .	00000000000000000002555566778	86
25.00	23 .	0000000000000000000002459	58
21.00	24 .	00000000000000000002448	33
12.00	25 .	0000000000000	12



E1

H1

M

H3

E2

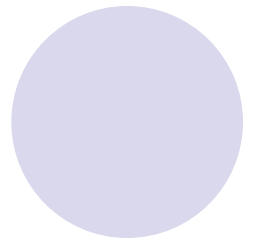
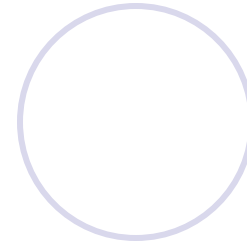
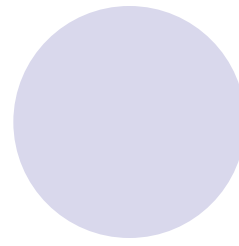
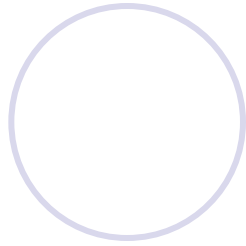
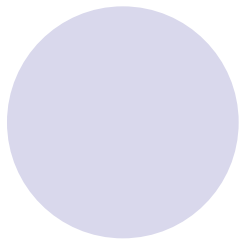
Box plot is defined by 5 characteristic values of data:

- Median M
- Quartiles H1 (25%) and H3 (75%) ,
- Smallest, Largest.



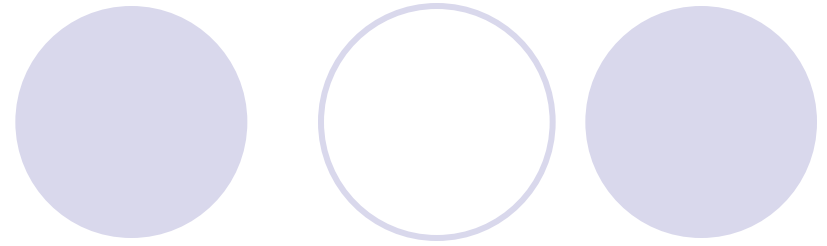
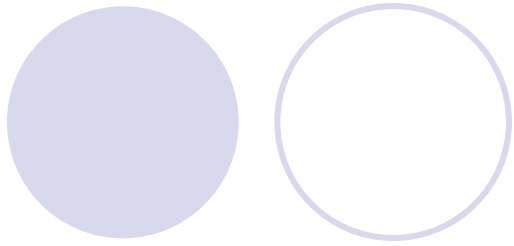
Five-Number Summaries and Box Plots

- Summary statistics and easy-to-draw graphs can be used to quickly summarize large quantities of data.
- Two tools that accomplish this are five-number summaries and box plots.



Five-Number Summary

1. Smallest Value
2. First Quartile
3. Median
4. Third Quartile
5. Largest Value



Box Plot

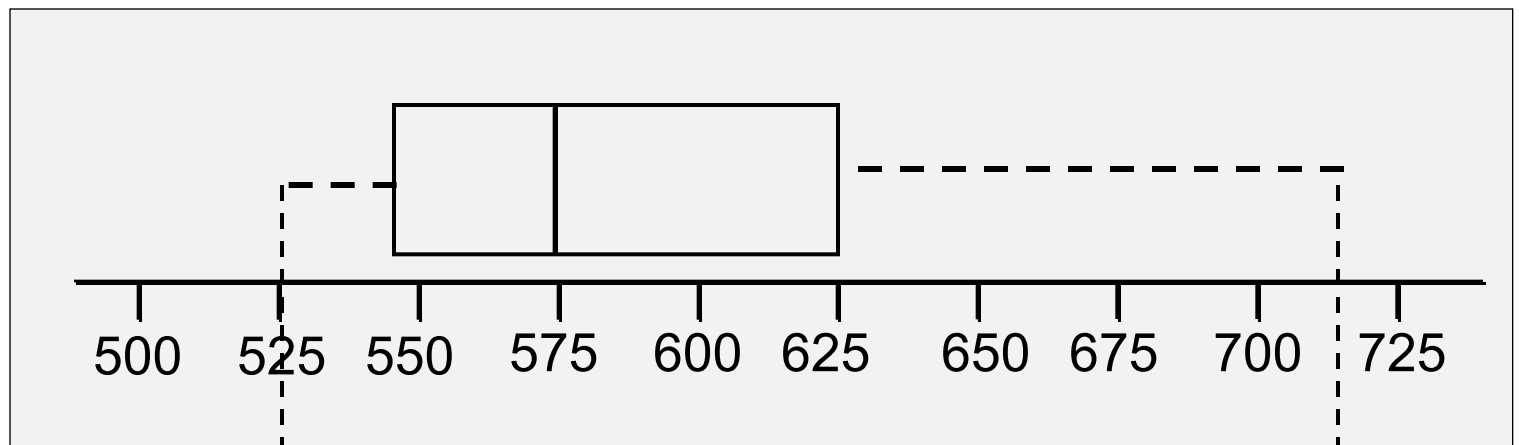
- A box plot is a graphical summary of data that is based on a five-number summary.
- A key to the development of a box plot is the computation of the median and the quartiles Q_1 and Q_3 .

Box Plot

- Limits are located (not drawn) using the interquartile range (IQR) multiplied by 1.5
- Data outside these limits are considered outliers.
- The locations of each outlier is shown with the symbol * .

Box Plot

- Example: Apartment Rents
- Whiskers (dashed lines) are drawn from the ends of the box to the smallest and largest data values inside the limits.

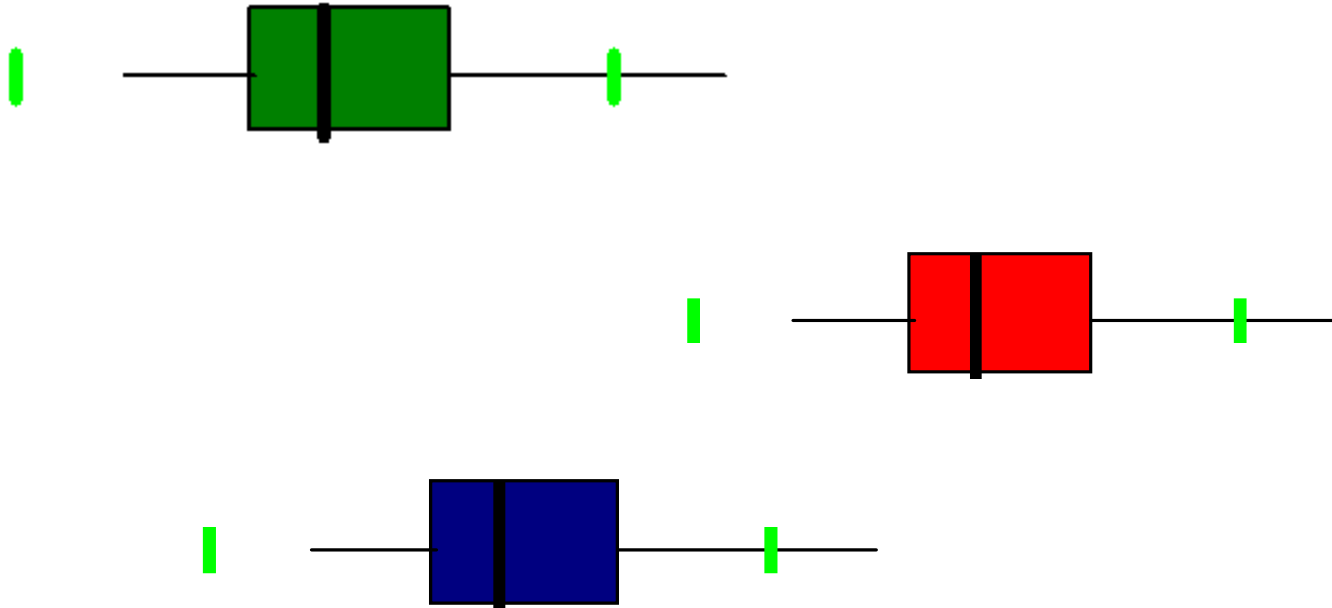


Smallest value
inside limits = 525

Largest value
inside limits = 715

2) Compare populations

Setting several box plots or stem-leaf plots each beside other, we can compare correspondent populations to see if there is any difference between populations



Using Excel to Describe Quantitative Variables

- MIN, MAX used to compute the extreme values.
- AVERAGE, MODE , MEDIAN used to compute the Mean, Mode and median parameters.
- VARP, VAR, STDEVP, STDEV used for the Variance and Standard Deviation parameters.
- PERCENTILE, QUARTILE used for the Percentile and Quartile parameters.
- HISTOGRAM used to draw the Histograms.



D. Describe relation between 2 quantitative variables

For primary describing relation between 2 quantitative variables we can use

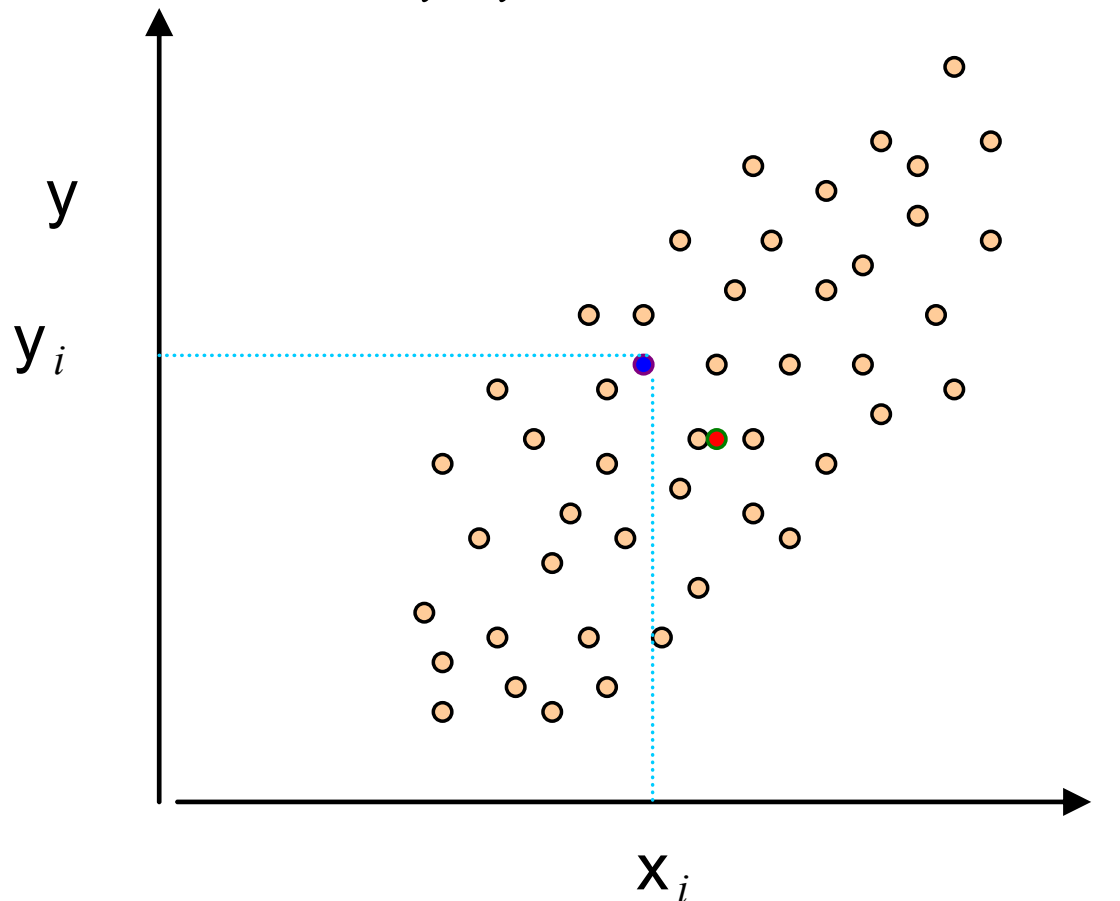
- Scatter plot,
- Covariance,
- Linear Correlation Coefficient.

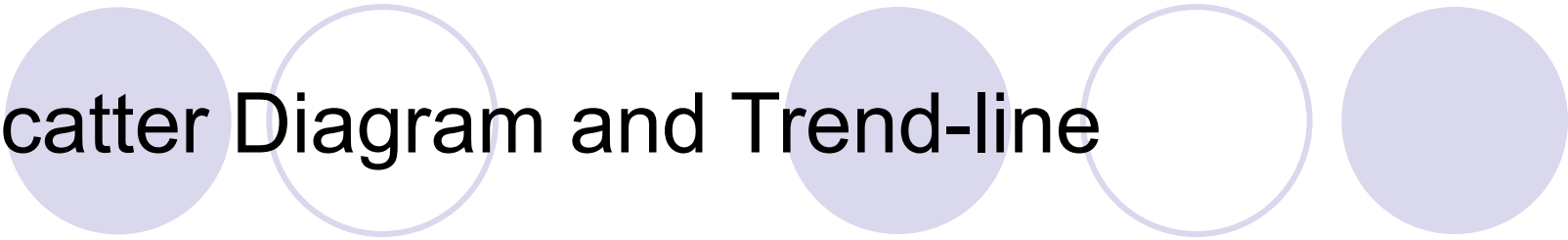
(a) Scatter plot

For quantitative variables $\mathbf{X}$, $\mathbf{Y}$ with sample

$$\mathbf{E} = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$$

Dot plot of sample $\mathbf{E}$ is performed by drawing n points with coordinates (x_i, y_i) :



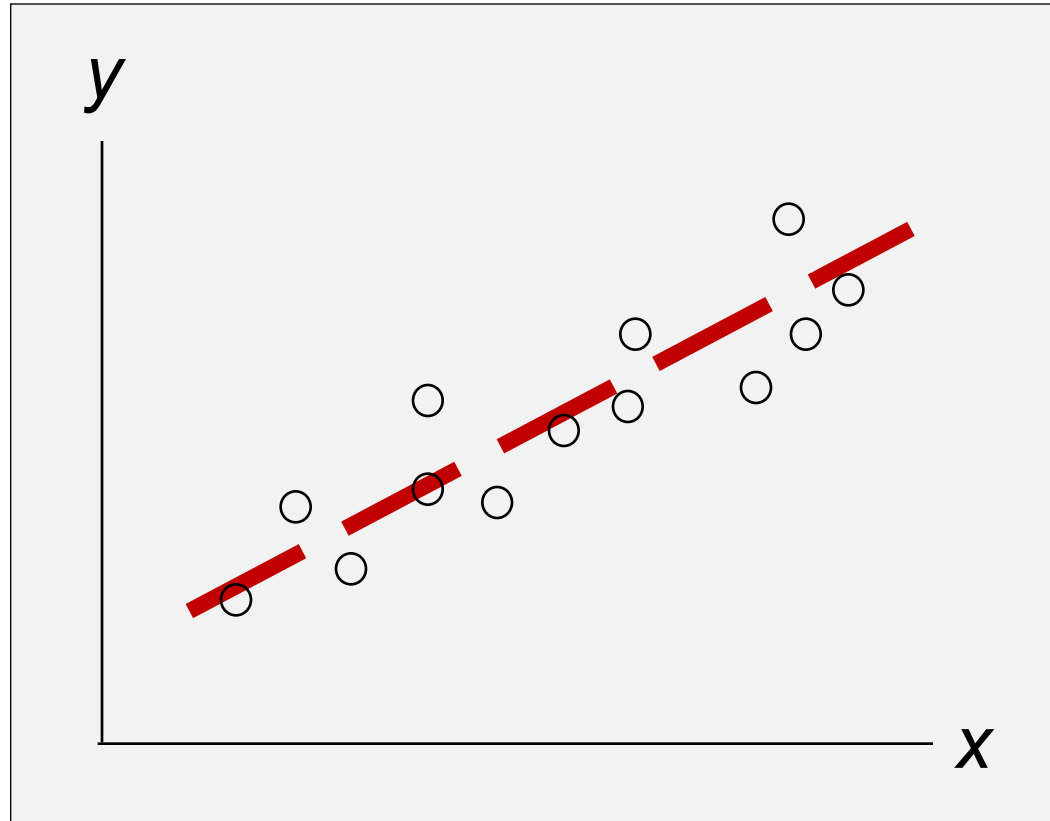


Scatter Diagram and Trend-line

- A scatter diagram is a graphical presentation of the relationship between two quantitative variables.
- The general pattern of the plotted points suggests the overall relationship between the variables.
- One variable is shown on the horizontal axis and the other variable is shown on the vertical axis.
- A trend-line provides an approximation of the relationship.

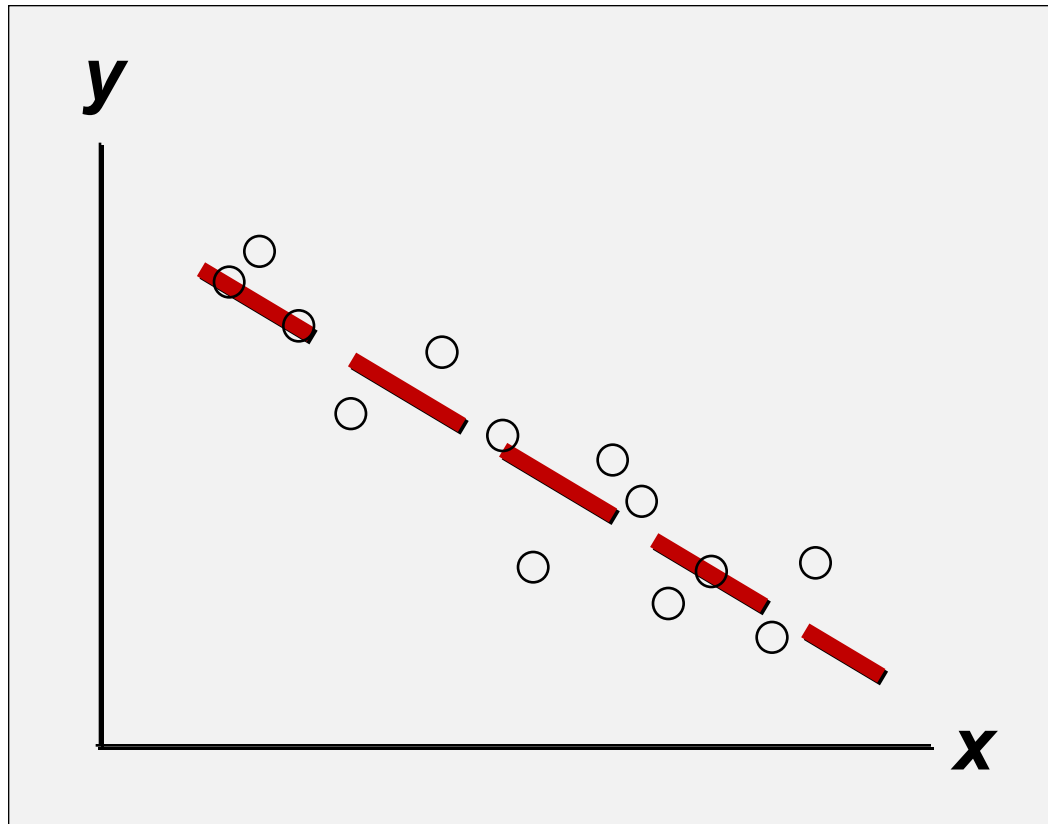
Scatter Diagram

- A Positive Relationship



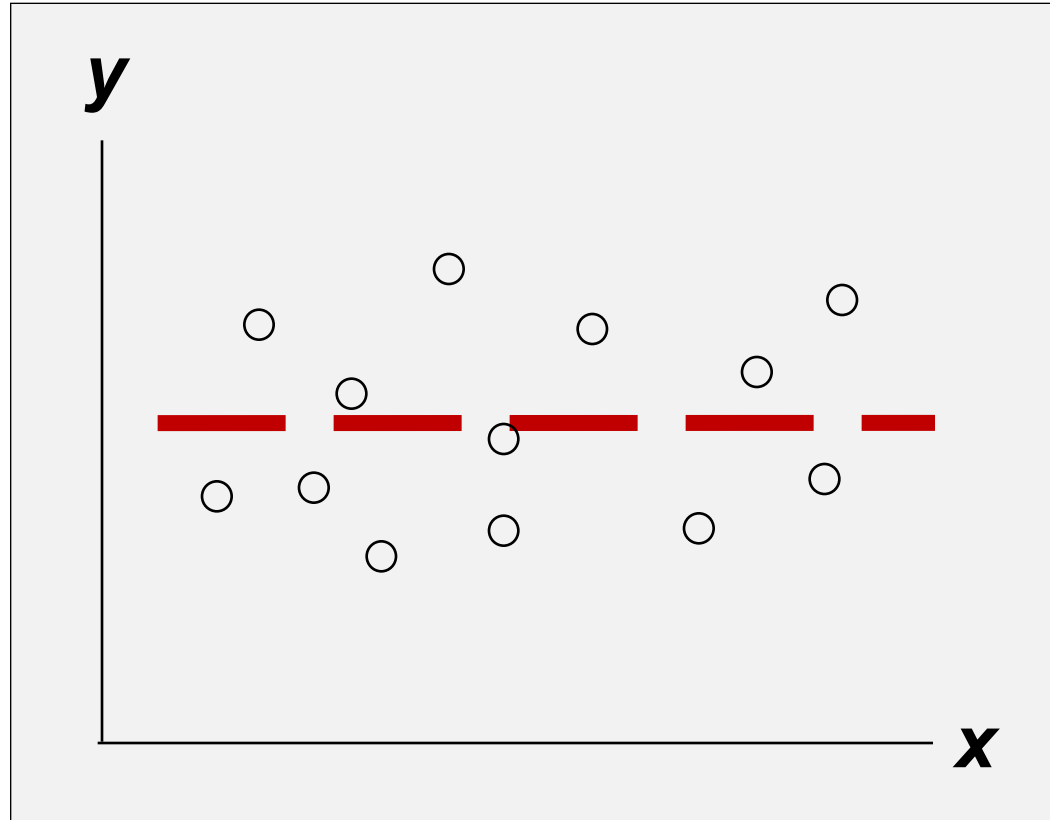
Scatter Diagram

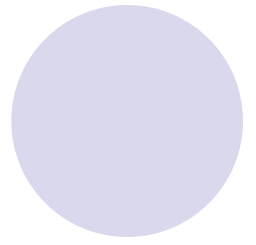
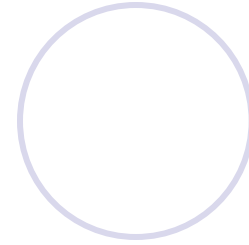
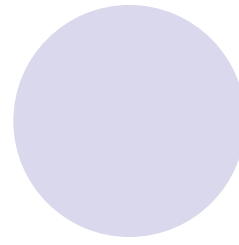
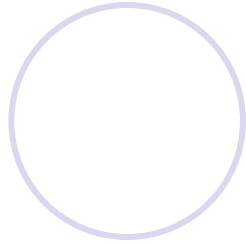
- A Negative Relationship



Scatter Diagram

- No Apparent Relationship





(a) Scatter plot providing two-dimensional picture of data represents distribution of data. In that plot we can see concentration area of data, see if there are some outliers, abnormal points, etc.

(b) Scatter plot can be used to compare several populations: draw several samples (differently coloured) on a common plot

Measures of Association Between Two Variables

- Often a manager or decision maker is interested in the relationship between two variables.
- Two descriptive measures of the relationship between two variables are covariance and correlation coefficient.

b) Covariance and linear correlation coefficient

(i) For presentation relationship between two quantitative variables we can use **covariance** between those variables

$$\text{Cov}(X, Y) = \frac{1}{n} \sum_{k=1}^n (x_k - \text{Mean}(X)) \cdot (y_k - \text{Mean}(Y)) .$$

Property of covariance

1) **Symmetric:** $\text{Cov}(X, Y) = \text{Cov}(Y, X)$

2) Depends on **measure scale** of X and Y :

$$\text{Cov}(a.X, b.Y) = a.b.\text{Cov}(Y, X)$$

Covariance

For samples:

$$s_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

For populations:

$$\sigma_{xy} = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{N}$$

Covariance

A decorative graphic at the top of the slide consists of two groups of circles. The group on the left has a solid light purple circle and an outlined light purple circle, with the word 'Covariance' overlaid on the solid circle. The group on the right has three circles: a solid light purple circle, an outlined light purple circle, and another solid light purple circle.

- The covariance is a measure of the linear association between two variables.
- Positive values indicate a positive relationship.
- Negative values indicate a negative relationship.

ii) *Linear correlation coefficient:*

$$r(X,Y) = \text{Cov}(X,Y) / (\sigma(\mathbf{X}) \cdot \sigma(\mathbf{Y})) .$$

Property:

1) **Symmetric:** $r(X,Y) = r(Y,X)$

2) **Not dependent** on measure scale of variables: For all numbers a, b different from 0 and $a.b > 0$ we have

$$r(aX, bY) = r(X, Y)$$

Correlation Coefficient

For
samples:

$$r_{xy} = \frac{s_{xy}}{s_x s_y}$$

For
populations:

$$\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$$



Linear correlation coefficient measures the linear dependence between two variables:

$$-1 \leq r(X,Y) \leq 1$$

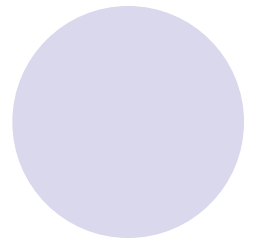
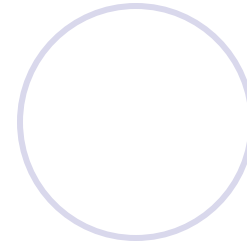
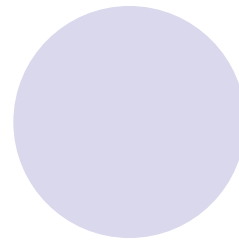
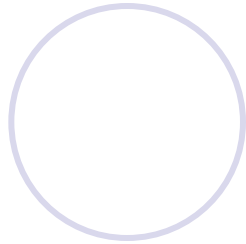
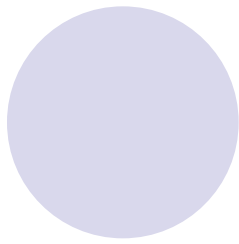
a) X and Y are completely linearly dependent:

$r(X,Y) = 1$ if and only if $Y = aX + b$ with $a > 0$,

$r(X,Y) = -1$ if and only if $Y = aX + b$ with $a < 0$.

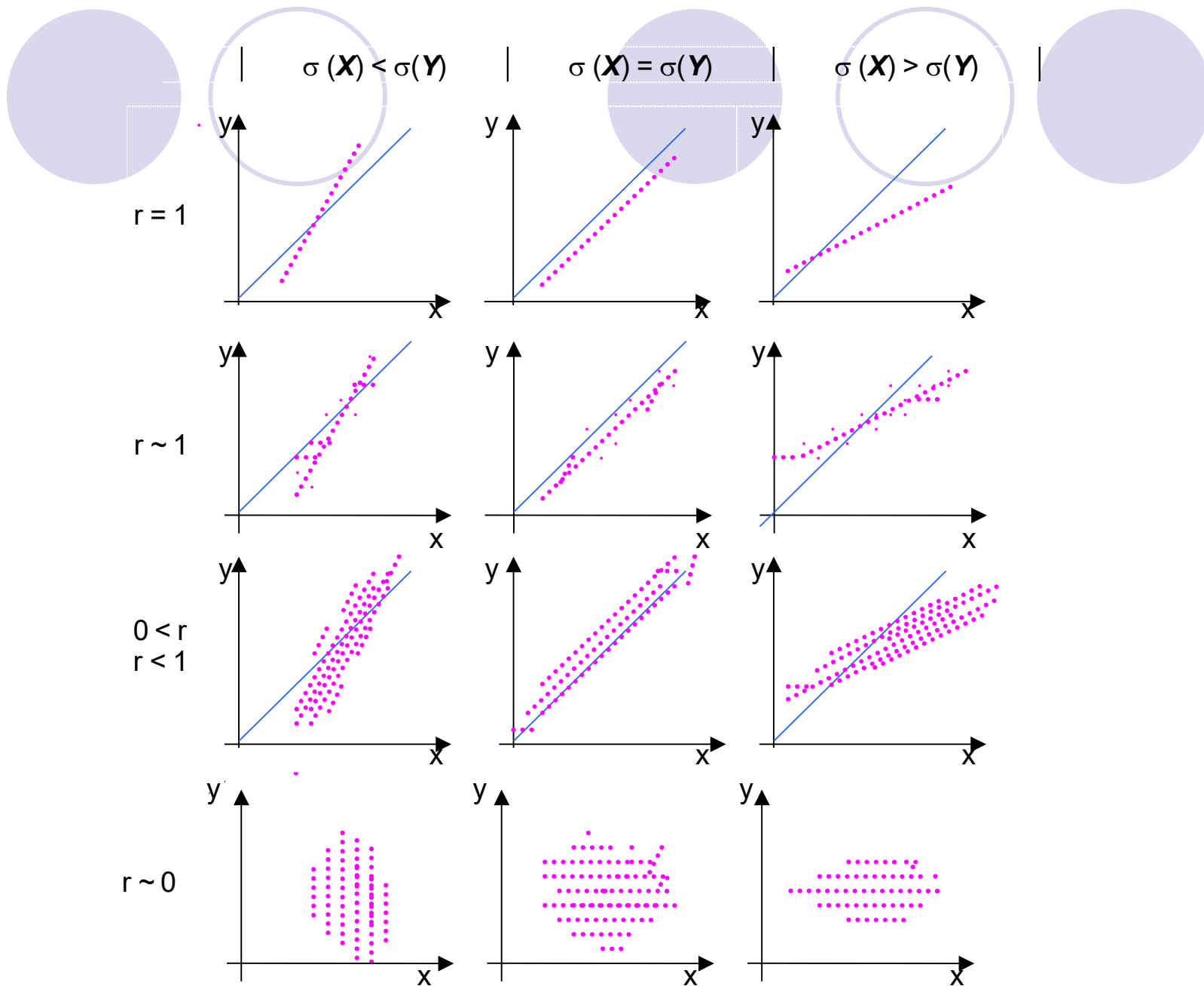
b) If $r(X,Y)$ close to 1 (or -1) then X and Y are very strongly related, can have some linear correlation,

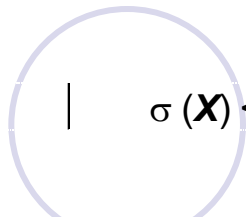
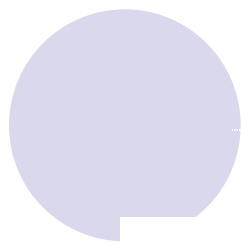
c) If $r(X,Y)$ close to 0 then X and Y are linearly independent, there is not linear relation between them.



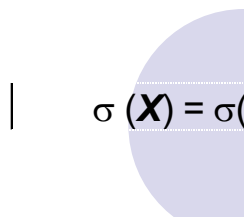
Correlation Coefficient

- The coefficient can take on values between -1 and +1.
- Values near -1 indicate a strong negative linear relationship.
- Values near +1 indicate a strong positive linear relationship.
- The closer the correlation is to zero, the weaker the relationship.

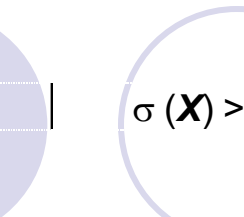




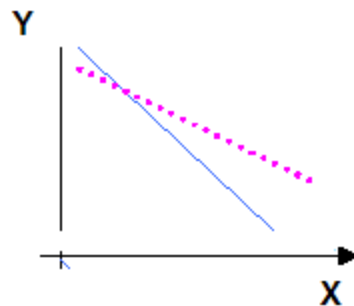
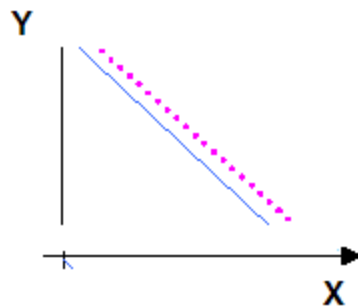
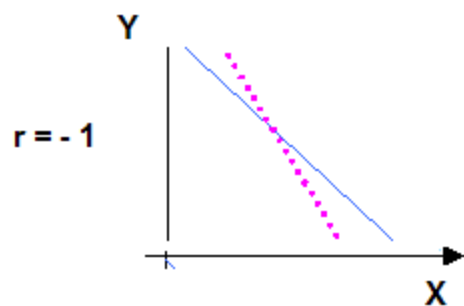
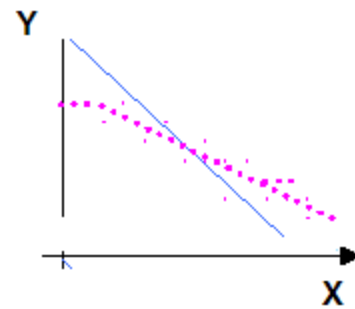
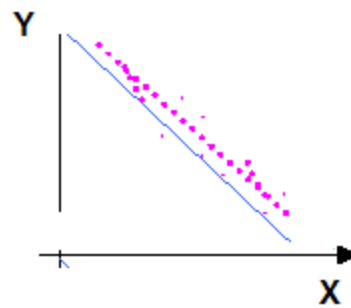
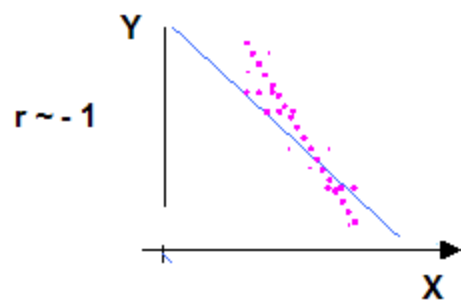
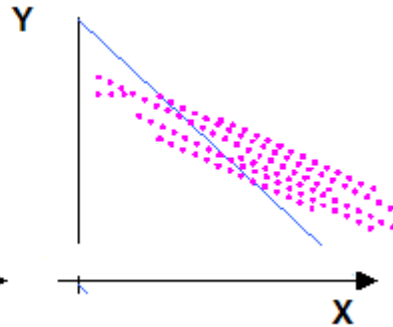
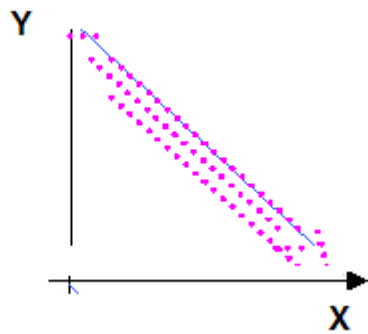
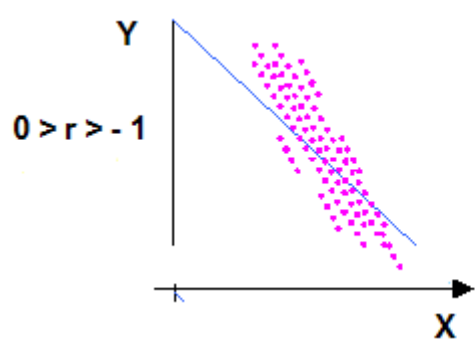
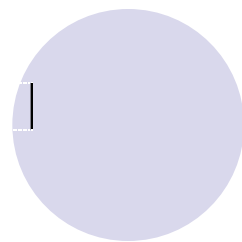
$$\sigma(X) < \sigma(Y)$$



$$\sigma(X) = \sigma(Y)$$



$$\sigma(X) > \sigma(Y)$$



Using Excel to Describe Quantitative Variables

- Excel has functions for computing the parameters of quantitative variables and draw graphs:
 - MIN, MAX, MED, MODE, QUARTILE, PERCENTILE used to compute the Minimum, Maximum, Median, Mode, Quartiles, Percentiles.
 - AVERAGE, AVERAGEIF, AVERAGEIFS used to compute the Mean Values.
 - VAR, VARP, STDEV, STDEVP used to compute the Sample Variance, Population Variance, Sample Standard Deviation, Population Standard Deviation.
 - COVAR, CORREL used to compute the Covariance, Correlation Coefficient.
 - HISTOGRAM, SCATTER used to draw the Histogram and Scatter Plot.